



Infor Mongoose Configuration Guide for Infor Operating Service

Release 2020.xx

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About this guide

This guide provides configuration and implementation information for the integration of Infor Mongoose with Infor Operating Service.

Use this guide when Mongoose and Infor OS are both installed on-premises.

Intended audience

This guide is intended for the system administrator or consultant who configures Mongoose for use with Infor OS.

Contacting Infor

If you have questions about Infor products, go to the Infor Xtreme Support.

If we update this document after the product release, we will post the new version on this Web site. We recommend that you check this Web site periodically for updated documentation.

If you have comments about Infor documentation, contact documentation@infor.com.

Contacting Infor

If you have questions about Infor products, go to Infor Concierge at <https://concierge.infor.com/> and create a support incident.

The latest documentation is available from docs.infor.com or from the Infor Support Portal. To access documentation on the Infor Support Portal, select **Search > Browse Documentation**. We recommend that you check this portal periodically for updated documentation.

If you have comments about Infor documentation, contact documentation@infor.com.

Chapter 1: Requirements

Ensure that all requirements are met.

Required products

Ensure that you have installed and configured these products:

- Infor Operating Service release 12 or later
The installation of Infor OS includes these products: Infor Ming.le, Infor ION, Infor Document Management, and ION Grid.
The server where you install Infor OS must be able to connect to the servers where you install Mongoose and other BOD-enabled products.
Security Assertion Markup Language (SAML) Session Provider is installed with Infor OS and must be configured correctly for your application.
See the *Infor Operating Service Installation Guide*.
- Mongoose 9.03 or later
See the *Infor Mongoose Installation Guide*.

Required information

Obtain this information before you begin the integration:

- Credentials to connect to the application interface, for example, database user, password, server name, port or instance.
- Name of the Mongoose instance or logical ID that is used to connect with Infor ION; tenant ID that is used to connect the product to Infor OS in Infor ION; and enterprise organization structure, that is, accounting entities.
See [Setting up BOD replication and logical IDs, tenants, accounting entities, and locations in Mongoose](#) on page 35 for information about how to set up the logical ID, tenant ID and accounting entities for your product.
- Mongoose sites that are used in the integration, and either the queue server name or the bootstrap site that is used to communicate with the sites through replication inboxes and outboxes.

- Names of physical locations where your Mongoose databases exist, if your company has multiple physical locations.
- An administrator account for Infor Ming.le and ION.
- Location or source of files to be imported into ION and Infor Ming.le for content.

Chapter 2: Configuration checklist

Follow this checklist to integrate this product with Infor ION:

Complete	Task	Reference
☐	Collect the documents listed in the next column from Infor Xtreme. These documents are required for this configuration.	<i>Infor Operating Service Administration Guide</i> <i>ION Desk User Guide</i>
☐	Understand the concepts of ION and BODs, and how these concepts relate to this product.	<i>ION Desk User Guide</i>
☐	Review the description of how this product interacts with Infor OS.	Integration with other products through ION Concepts and definitions specific to this configuration on page 16
☐	Configure your product for Single Sign On: • Set up your product to use Single Sign On. • Install Windows Identity Foundation. • Set up the Web client with Active Directory Federation Services (AD FS) access. • Edit the web.config file. • Create a relying party trust for Infor Ming.le and Mongoose. • Configure automatic login. • Configure Active Directory authentication groups. • Import a security certificate for each user computer. • Set the time-out option for all applications.	Configuring your product for Single Sign On on page 18.

Complete	Task	Reference
☐	Add your application in Infor Ming.le: - Set up compatibility for Internet Explorer browsers. - Add the application. - Add accounting entities. - Add locations.	Adding your application in Infor Ming.le on page 23
☐	Configure user access to your product through Infor Ming.le: - Set up BODs to pass user and role information. - Configure user access to your product in Infor Ming.le. - Configure system administration access to your product through Infor Ming.le. - Set up roles in Infor Ming.le. - Send the Person ID from your product to Infor Ming.le. - Set up distribution groups.	Configuring user access to your product through Infor Ming.le on page 27
☐	Configure your product and ION to send and receive BODs - Configure this product for ION. - Configure a connection point for this product. - Publish BODs. - Verify the ION configuration.	Configuring your product and ION to send and receive BODs on page 34.
☐	Configure drillbacks to your product.	Configuring drillbacks to your product on page 43
☐	Configure context and utility apps that are used with your product.	Configuring context apps and utility apps that are used with your product on page 44
☐	Configure Homepages for your product.	Configuring Homepages for your product on page 48
☐	Optionally, configure workflows and ION messages for your product. If you plan to use the sample workflow content, follow these steps: - Install the content. - Configure drillback views for Workflow modeler.	Configuring workflows and ION messages for your product on page 49

Complete	Task	Reference
☐	Configure your product to work with Infor Document Management: • Import configurations and types. • Enable the Related Information context app for your product.	Configuring your product to work with Infor Document Management on page 56
☐	Set up ION APIs for your product.	
☐	Verify the configuration of this product with Infor OS.	“Verifying the configuration”, <i>Infor Operating Service Administration Guide</i>

Chapter 3: Overview

Before you complete the configuration tasks, you must understand how this product is used with Infor OS. You should also be familiar with a list of related concepts and definitions.

Infor ION

Infor ION is an interoperability and business process management platform designed to integrate applications, people, and processes to run your business. Infor ION enables Infor and customer activities to configure a routed infrastructure, set up workflows, design and activate business event monitors, and manage the tasks and alerts through Infor Ming.le.

Infor ION has these components:

- With ION Connect, you can establish connections between applications, which can either be Infor applications or third-party applications. You can configure ION Connect through the ION Desk.
- ION Desk is an intuitive, browser-based interface used to easily configure and monitor the ION Service.

In ION Desk, you can model document flows between applications. Such flows can represent a business process. More technical flows can also be defined. For example, mapping data from a third party application to a standard business object document as used by an Infor application. You can also use filtering or content-based routing.

From ION Desk you can deploy your models to ION Service by activating them. ION Service will then handle your documents in accordance to your activated models. ION Desk also provides management screens to monitor the behavior of ION Service and to help in troubleshooting when needed.

- In ION Event Management you can monitor business events and create alerts when exceptions occur.

The start point of monitoring are the business object documents (BOD) that are published by applications when business events occur. Monitors are processes that run in the Event Management engine and evaluate these documents by applying pre-defined business rules.

There is a predefined set of business rules that the monitoring engine can apply: comparison conditions, value change conditions, and document overdue timer based rules. When exceptions are detected, these are reported as alerts to the business users. The distribution list of alerts is part of the monitor definition.

- ION Workflow is used to model and run business processes such as authorizations, work distribution, or task-driven workflows across several components.

For authorizations, you can use an approval workflow to approve a purchase order or to sign off a new item. Such workflows are triggered by the creation of a new item or the submission of a new order.

In the area of work distribution, the workflow ensures that all tasks for a certain event are being distributed and executed by different users. The advantage of using the ION Workflow is that you can model how these tasks are performed, for example sequential or in parallel.

- ION Pulse is the component that ensures the communication to the end user. ION Pulse manages the Alerts, Tasks and Notifications and distributes them to users. When users take actions in Infor Ming.le, ION Pulse records these actions and returns the reply to the engine that generated the user activity. ION Pulse is also the component that sends email notifications for Tasks, Alerts, and Notifications.

For more information about these components, see the *Infor ION Desk User Guide*.

To understand common ION concepts such as BOD nouns and verbs, document flows and workflows, listeners and connection points, see the ION Connect Concepts chapter in the *ION Desk User Guide*.

Integration with other products through ION

An outbound operation typically begins when a user performs an action in Mongoose that requires a data exchange with another ION-enabled product. A Business Object Document (BOD) XML message is generated by Mongoose and placed in an area designated as the Mongoose message outbox. At scheduled intervals, ION connects to the outbox and retrieves the BODs from it.

In ION, you create application connection points, which define the connections between ION and a product that can send and receive BODs. ION Connect routes BODs according to the document flows between Mongoose and other ION-enabled products.

The document flows between Mongoose and other products represent the business relationship between the databases. You use the ION Document Flow Modeler page to define these document flows.

If a document flow is defined from Mongoose to another product for a particular BOD, then at specified intervals, ION places the outbound BOD from Mongoose in the other product's designated message inbox. Products are responsible for validating and incorporating the data in inbound BODs according to their rules.

If a flow is defined from another product to Mongoose, ION retrieves BODs from the sending product's message outbox and delivers them to Mongoose's message inbox for processing. Mongoose retrieves, validates, and processes the BODs.

ION Grid

ION Grid is an application server, or a set of connected servers, that run many different applications. A grid application is a logical grouping of one or more application modules. If an application can run in more than one node, then there are multiple instances of the application.

See the *ION Grid Administration Guide* for administration tasks such as monitoring, configuring, or managing the grid.

Infor Ming.le

Infor Ming.le is an application framework that provides a common user interface for integrated Infor applications. Infor Ming.le provides a centralized platform for employee collaboration where information is shared and conversations are organized across Infor Ming.le.

The Infor Ming.le interface includes a top navigation panel with an application panel to activate the display of Infor applications.

Drillbacks

Infor Ming.le users can share screens and business data in some ION-enabled products that are integrated with Mongoose. The screens and business data can include embedded drillback links to Mongoose. If a screen or business data is associated with a BOD that was sent from Mongoose, and the screen or data has an associated drillback link, then users can click the link to view the screen that has the supporting information for the source transactions.

A drillback link can be defined for each type of BOD noun, for example, SalesOrder, or ItemMaster. The link is configured to open a specific Mongoose screen for that noun. Not every BOD noun has an associated drillback view in Mongoose. See [Available drillbacks](#) on page 43.

To set up the drillbacks, see [Importing and configuring application drillbacks](#) on page 43.

Context applications

On the right side of the Infor Ming.le page is a collapsible panel that hosts a series of Infor Ming.le context applications. The context applications are mini applications that provide information at a glance. The information in the applications is context-sensitive, which means that the values or content in the application depend on the current context (form or field) that is shown in the product application panel in Infor Ming.le.

See [Configuring context apps and utility apps that are used with your product](#) on page 44.

Homepages

Homepages are a flexible and configurable way to present information and activities to users. Administrators can build pages for various groups of users, roles, departments, or business functions. You can select from the library of available Infor widgets or build your own.

Infor Document Management

Infor Document Management enables customers to manage business documents related to application objects and make them available to the user in Infor Ming.le. It provides a versatile solution for storing, sharing, reusing, and archiving digital content. This product reduces the time and cost for searching documents. Infor Document Management is secure and provides granular access control, enabling users to add security and permissions down to individual pieces of the document.

Infor Document Management also provides a context application that can be used to show context-based documents next to the application.

Concepts and definitions specific to this configuration

To configure the tenant, logical ID, accounting entity and location correctly, you must understand these terms and how they are defined in this product. Together, these terms determine where inbound BOD information is processed, or the outbound instance and location to which the BOD information should be associated.

Tenant ID

The tenant is a container for accounting entities and locations and is required in each BOD. The value of the tenant must be the same in all of the products that exchange BODs. Data is not shared or accessible between tenants.

See the information about Using Tenants in the *Infor ION Desk User Guide*.

For an on-premises configuration, verify that all of the other products are using the default value of **infor** for the tenant ID, and if so, use that as your tenant ID for Mongoose.

Accounting entity

An accounting entity usually represents a legal or business entity that owns its general ledger. Every transaction belongs to only one accounting entity. Accounting entity can also be defined as the owner of certain master data among the enterprise.

For Mongoose, the site is used as the accounting entity.

Location

A location is the physical location that is associated with data or transactions. The location can be, for example, a warehouse, a manufacturing location, a project location, or an office. Locations are published in the Location BOD. A location is required for all transactional BODs and is usually the location from which a transaction or record is generated.

Logical ID

In ION, the logical ID is the instance of the product that sends or receives BODs. One logical ID is defined for each product instance. This logical ID is defined in the Mongoose Sites form. You must provide the logical ID when you define the ION connection point for on-premises products.

Note: In Mongoose, the logical ID refers to a Mongoose Site. In some Mongoose-based applications, the logical ID can be Site/Entities.

In Infor Ming.le, each application is assigned a logical ID. Infor Ming.le supports only one logical ID per application.

Because Mongoose supports multiple instances, you must use the Logical ID Fallback table in Infor Ming.le to set up the relationship between the instances. For example, the Infor LN is assigned the logical id lid://infor.ln.01 in Infor Ming.le. Infor Ming.le uses this logical ID to find the target for drillback or context messages. However, LN users can select different companies (instances) through one Web UI.

Each company within LN has a logical ID. The LN application can access these companies:

- Company 1 (lid://infor.ln.01), which is recognized by Infor Ming.le
- Company 2 (lid://infor.ln.02) and Company 3 (lid://infor.ln.03), which Infor Ming.le does not recognize

Use the Logical ID Fallback table to associate additional instances of LN with instance 01. Then, when a drillback is sent to Infor Ming.le for lid://infor.ln.02, Infor Ming.le uses the information in the Logical ID Fallback table to determine that instance 02 is related to 01. Infor Ming.le passes the drillback information to LN instance 01 and the request is handled by LN.

For more information, see the *Infor Operating Service Administration Guide*.

Chapter 4: Configuring your product for Single Sign On

Single sign on allows users to use one set of credentials to log in to all Infor applications.

Mongoose is set up for Single Sign On (SSO) in Infor Ming.le through the use of SAML authentication, Microsoft AD FS, and Integrated Windows Authentication. This requires an environment where a Windows domain trust between the client systems and the AD FS host is configured.

See the *Infor Operating Service Installation Guide* for information about the client Single Sign On experience and about the SAML session provider.

Setting up your product to use Single Sign On

Use these steps to set up Single Sign On between the Infor Ming.le Portal and Infor Mongoose.

Installing Windows Identity Foundation

Windows Identity Foundation 3.5 must be installed on the AD FS server and the Mongoose application server. Parts of this are installed during the Mongoose installation, but the full feature must be installed in order to use SSO.

See the *Infor Mongoose Installation Guide* for information about setting up Mongoose to work with AD FS.

Setting up the web client with AD FS access

The Mongoose web client can run as an application inside the Infor Ming.le shell. Be aware that there are some differences between the Mongoose web client and the smart client. These differences are documented in the Mongoose online help.

Ensure that these web client requirements are met:

- 1 Install a Mongoose web client on a server, as described in the *Infor Mongoose Installation Guide*.

- 2 For Mongoose installed on-premises, in order to access online help from Mongoose in Infor Ming.le, install the application server where the help files reside on the same physical server where the web client is installed.
- 3 Follow the steps to set up web rendering with AD FS as described in the *Infor Mongoose Installation Guide*.
- 4 In the Mongoose Configuration Wizard, select **AD FS** as the authentication type.
- 5 In AD FS, create a relying party trust and give it the appropriate claims rule.
- 6 In AD FS, add all of your Mongoose users, which will allow automatic login to Mongoose through Infor Ming.le.
- 7 Verify that you can log in to the web client configured with AD FS.
Take note of the URL that you use to log in, because you will need it for a later step.

Editing URLs in the web.config files

For on-premises installation, the Mongoose web client only renders URLs that are on the same server as the web client.

Edit the web.config files in both of these locations under the inetpub/wwwroot folder on the application server:

- WSWebClient
- IDORequestService

To accept URLs from other locations, remove these associated tags in the web.config files:

- Remove the element `<add name="X-Frame-Options" value="SAMEORIGIN"/>`
- Remove the element `<add name="Content-Security-Policy" value="frame-ancestors 'self';"/>`

Alternatively you can edit these entries to add servers for more security. For example, add a value of `"frame-ancestors 'self' https://mingleserverurl.com"`. The web.config file can also accept wildcards such as `"frame-ancestors 'self' https://*.infor.com"`.

Creating a relying party trust for Infor Ming.le and Mongoose

This task uses a Power Shell script to create a relying party that associates Infor Ming.le with AD FS for Single Sign On.

This Power Shell script is similar to the Power Shell scripts that you ran in the post-installation steps of Infor OS installation. See the *Infor Operating Service Installation Guide*.

- 1 In Infor OS Manager, open the Applications section.
- 2 Add an application and specify this information:

Application Name and Description

Specify a name of your choice that identifies the product.

Application Type

Select **MONGOOSE (Mongoose)**

Relying Party Identifier

Specify a unique ID in this format, where *URN* is a name of your choice that identifies the product and party ID:

Urn: *URN* :Mongoose

For example, Urn: **Infor_Mongoose** :Mongoose.

SSO URL and SLO URL

Specify the web client URL that has an .aspx extension.

Signing Certificate

Leave this field as **(none)**.

- 3 Save your changes.
- 4 Click the arrow next to the new application and specify a location to download a Power Shell script that is generated from this information.
- 5 Copy the Power Shell script to the AD FS server.
- 6 Open the Windows PowerShell as an administrator.
- 7 Run the `Set-Execution Policy Unrestricted` command and specify **y** to confirm the Execution Policy.
- 8 Press Enter.
- 9 Run the PowerShell script that you downloaded.
- 10 Configure Forms Authentication:
 - a Open AD FS Management and click **Authentication Policies**.
 - b Click **Edit** in the Primary Authentication section.
 - c Select **Forms Authentication for Extranet and Intranet**, clear **Windows Authentication for Intranet**, and click **Apply**.

Configuring automatic login

Use this procedure to allow users to automatically log in to this product within Infor Ming.le:

When users first access Mongoose through Infor Ming.le, they see the Mongoose login dialog. On that dialog, they must select the configuration they want to use, and select **Use Workstation Logon**. From then on, they are automatically logged into Mongoose when they select Mongoose from the **App Menu** in Infor Ming.le.

After you add your users in Infor Ming.le and the information is sent through BODs to Mongoose, you will see that the user name (email address) for each user is shown in the Mongoose Users form for each user, in the **Workstation Domain/ID** field.

Configuring Active Directory authentication groups

For access to the Infor Ming.le and Mongoose sessions, you must associate all users' domain accounts with these predetermined Active Directory security groups:

Infor application	Active Directory security group
Infor Ming.le	Infor
Mongoose	Mongoose Users

Create these security groups in Active Directory outside of Infor Ming.le. See the Microsoft documentation about Active Directory for information about how to configure user accounts with security groups. Also see the *Infor Mongoose Installation Guide* for information about how to configure Mongoose for use with Active Directory.

Importing a security certificate for each user computer

To allow AD FS Single Sign On to Infor products, you must obtain a certificate that your users can add to their computers by using the Microsoft Management Console.

To import the certificate to a user's computer:

- 1 From the Windows menu, select **Run** and enter **mmc**.
- 2 Select **File > Add/Remove Snap-in**.
- 3 Select **Certificate** and click **Add**.
- 4 Select **Computer Account** and click **Next**.
- 5 Expand **Certificates**.
- 6 Right-click **Trusted Root Certificate Authorities**.
- 7 Select **All Tasks > Import** and import the certificate file.

Chapter 5: Single Sign Out

When a user signs out of a portal such as Infor Ming.le, the user is automatically logged out of all applications, context applications, and widgets that were opened by the user through the portal.

Setting the time-out option for all applications

Use the Time-out option in Infor Ming.le to configure the idle session time-out value for all the Infor applications that are accessed through the Infor Ming.le portal. This time-out value applies to all users and applications in Infor Ming.le. User activity is updated periodically by each application through a common cookie in the root DNS domain. If no user activity is logged within the period specified in the time-out value, the user is automatically logged out of all applications.

- 1 Select **Time-out** from the Infor Ming.le **Security Administration** menu.
- 2 Configure the idle session time-out value to use for all Infor applications that are accessed through the Infor Ming.le portal. The time is measured in minutes, and you can specify a number from **10** to **470**.
- 3 Configure a session timeout value for the Mongoose web client that is at least as long as the Infor Ming.le timeout value. Set the timeout value in the web.config file located in the WSWebClient IIS application folder on the Mongoose application/web server. For example, to set the Mongoose web client session timeout to 60 minutes, use this element and attribute value:

```
<sessionState mode="InProc" timeout="60">  
</sessionState>
```

Chapter 6: Adding your application in Infor Ming.le

You must add your application in Infor Ming.le.

Setting up compatibility for Internet Explorer browsers

Follow these steps only if you are using Internet Explorer as your browser.

Some Infor Ming.le and Mongoose versions of Internet Explorer might be incompatible. If so, a message is displayed to indicate that your browser is not supported. In that case, before you can run Mongoose in Infor Ming.le, you must change your compatibility view settings in your browser.

- 1 In the browser, click **Tools > Compatibility View Settings**.
- 2 Clear the **Display intranet sites in compatibility view** option.

Adding the application

Perform these steps to add Mongoose as an application that can run in Infor Ming.le:

- 1 In Infor Ming.le, add a new application.
See the *Infor Operating Service Administration Guide*.

- 2 Use this information to complete these fields:

Application Type

Specify **Infor Application**.

[In the cloud, options are **Infor Provisioned** or **Infor Non-provisioned**.]

Application Name

Select your application name and version from the list.

Logical ID

The first part of the logical ID is set automatically based on the application name. Specify the rest of the logical ID for the Mongoose instance associated with this instance of Infor Ming.le, for example, the site name.

Application Icon

Click **Choose Icon**.

Each application has a default icon. You can only select the color which you want to apply to the icon.

Use Https

Select this option if SSL is enabled for the Mongoose user interface. If Mongoose is using SSL, ensure that you specify the SSL-related port number in the **Port #** field.

Host Name

Specify the name of the Mongoose web/utility server. This host name is also used for drillbacks to the application and for accessing the online help and documents from the Documentation context application.

Also, if you use a fully-qualified name, for example, server.infor.com, in one instance, you must use it for all instances.

Note: The host name entered in Infor Ming.le must exactly match both the web server referenced in the Application URL specified in the Web rendering section of the Infor Mongoose Configuration Wizard and the host name specified on the server's security certificate. The URL names are case-sensitive.

Port #

Specify the port used by the Mongoose web/application server. Typically this is **80** for HTTP, and **443** for HTTPS.

Context

If any context is defined in the Infor Registry, it is shown as the default context when you select the **Application Name**.

Default Tenant

We recommend that you specify **Infor**.

3 In the Application Details pane, review the default values in these tabs:

- **Deployment Information** - The connection information you specified for this application is displayed.
- **Permissions** - Two default roles are created, Infor-SystemAdministrator and MingleAdministrator. Additional roles are automatically generated that match the default authorization groups in Mongoose.

Later in the configuration process, you will use the **Permissions** tab to configure user access to Mongoose through Infor Ming.le.

- **Context/Utility Apps** - The default context apps that are included with your product are listed on this tab. See [Supported context and utility applications that are delivered with Infor Ming.le](#) on page 45.

You can add other context or utility apps here, including context apps that are unique to your product. See [Context applications that are delivered with your product](#) on page 46.

By default, all the required applications are enabled and optional applications are disabled. You can enable or disable context applications at any time.

- **Custom Parameters** - Default custom parameters, if any exist, are displayed. You cannot edit the default name of the custom parameter.

- **Logical ID Fallback** - By default, no logical ID fallbacks are displayed.

Default values are generated when the application is created. Do not change the default values.

Adding accounting entities

In Infor Ming.le, you must specify the accounting entities associated with Mongoose and the users and groups who are part of that accounting entity.

You can import accounting entities and the associated users and groups from a CSV file that includes these headers:

- Name
- Description
- UserIdentifier
- GroupId

The **Name** and **Description** fields are required in the header row.

The application is the System of Record (SOR) for accounting entities. The accounting entities are stored in Infor Ming.le and are set through BODs.

Adding locations

In Infor Ming.le you must specify the locations associated with Mongoose and the users and groups who are part of that location.

The Mongoose application is the System of Record (SOR) for locations. The locations are stored in Infor Ming.le and are set through BODs.

Setting up Mongoose web part URL

Web parts are instances of the Mongoose web client running in form-only mode in an iframe. When constructing the URL, the administrator must use one of these access points:

Guest account	<code>http://domain/WSWebClient/AppApi/Guest/config?page=formonly&form=formname</code>
Native Mongoose security	<code>http://domain/wswebclient/session/open/config/user/pswd?page=formonly&form=formname</code>

SSO	http://domain/wswebclient/session/open/config?page=formonly&form=formname
-----	---

Chapter 7: Configuring user access to your product through Infor Ming.le

Users, roles, and person IDs are defined and shared between Infor Ming.le and all of the other applications that run through the Infor Ming.le portal.

Users and roles

Because Infor Ming.le is the System of Record for users, you must set up users in Infor Ming.le and then synchronize the users through BODs to other applications in the Infor Ming.le portal. If your product requires additional user setup beyond the configuration that is done in Infor Ming.le, then after the users are synchronized to your product, you can continue the user setup in your product.

You can perform these user management tasks in Infor Ming.le:

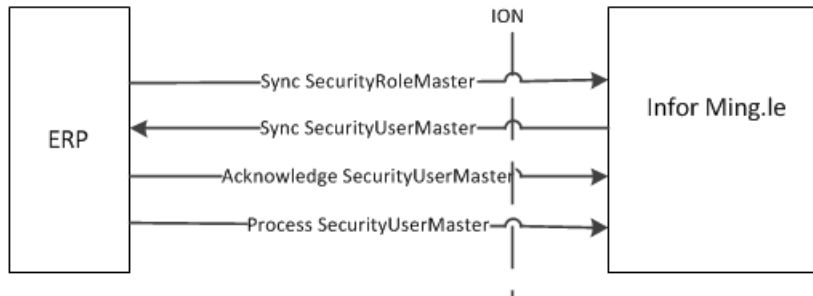
- Create users, including manual import of users
- Update users
- Delete users
- Reset passwords
- Monitor recent user activity and mobile sessions
- Manage sessions

When you create or update a user in Infor Ming.le, the information associated with that user is used to automatically set up some user information in Mongoose. This association is handled through Sync SecurityUser BODs that are passed through Infor ION.

By default, Mongoose is the System of Record for user roles, because the types of roles and the names of roles are dependent on the requirements of Mongoose. After the roles are set up in Mongoose, they are synchronized to Infor Ming.le through the Sync SecurityRoleMaster BOD, so that Infor Ming.le is aware of all possible roles. The ability of security roles to perform specific operations and access specific data is enforced by Mongoose, not Infor Ming.le.

Setting up BODs to pass user and role information

Ensure that connection points are set up and a document flow is activated in ION Desk so that these BODs are sent between Mongoose and Infor Ming.le:



- 1 In ION Desk, verify that a document flow to an Infor Ming.le connection point exists.
- 2 You should have already set up the connection point when you completed the post-installation steps when you installed Infor OS. See the *Infor Operating Service Installation Guide*.
- 3 Edit the document flow between the Mongoose connection point and the Infor Ming.le connection point.
See [Configuring a document flow between products](#) on page 40.
- 4 Add the BODs shown in the flowchart to the document flow, in the appropriate direction.
- 5 Activate the document flow.

For information about these BODs, see these topics:

- [BODs that send authorization groups \(roles\) from your product to Infor Ming.le](#) on page 28
- [BODs that send user information from Infor Ming.le to your product](#) on page 29
- [BODs that send user information from your product to Infor Ming.le](#) on page 29

BODs that send authorization groups (roles) from your product to Infor Ming.le

Default authorization groups from Mongoose are preconfigured in Infor Ming.le when you add the Mongoose product as an application in Infor Ming.le.

When you create or edit an authorization group in Mongoose, a Sync SecurityRoleMaster BOD is generated. This BOD is defined at the tenant level. The information from this BOD is used in Infor Ming.le to automatically create a matching security role.

If you create a security role in Infor Ming.le that does not exist in Mongoose, you must manually create a matching role in Mongoose. If you delete a role in either Mongoose or Infor Ming.le, you must also delete the group or role in the other application.

BODs that send user information from Infor Ming.le to your product

When you add a user in the Infor Ming.le portal, a SecurityUserMaster BOD is sent through ION to your product at the specified intervals, where a matching user record is created or updated. This BOD contains the user name, login, creation date, status, description, email address, and the roles, or groups, to which the user is assigned.

When Mongoose receives the BOD, these actions occur:

- For a new user that does not exist in Mongoose:
 - A new user record is added
 - The group authorizations for that user are added.
 - A primary email type description is added that contains the user's email address.
 - The user's Given Name and Surname values from Infor Ming.le are combined into the Mongoose User Description field, separated by a space.
- For an existing user in Mongoose, if the existing user ID has not changed:
 - The existing user group authorizations are deleted.
 - The user status, user descriptions, primary email address, and user group authorizations are updated.
- For an existing user in Mongoose, if the existing user ID has changed:
 - The record for the user with the matching Workstation Login is set to a status of disabled, and the Workstation Login for the user is cleared.
 - A new user record is created, copying the user information from the disabled record. Any existing user group authorizations in Mongoose are deleted, and then the groups from the BOD are added. The status, email, and description are updated if they have changed.
 - A Sync SecurityUserMasterAcknowledge BOD is sent back to Infor Ming.le.

BODS that send user information from your product to Infor Ming.le

When any of this information is manually added or updated in Mongoose for a user, a Process SecurityUserMaster BOD is published by Mongoose:

- User login status
- User Description - A value is required in this field if you are passing user information from Mongoose to Infor Ming.le through the Process SecurityUserMaster BOD. The field must contain values separated by a space. The first value becomes the Given Name in Infor Ming.le. Everything after the space becomes the Surname in Infor Ming.le. For example, if the Mongoose User Description is Tom Van Winkle, Tom is the Given Name and Van Winkle is the Surname in Infor Ming.le.
- Workstation login
- Primary email address - A value is required in this field if you are passing user information from Mongoose to Infor Ming.le through the Process SecurityUserMaster BOD.
- Group authorizations

However, if the user information was changed in Mongoose because of an incoming Sync SecurityUserMaster BOD from Infor Ming.le, the Process SecurityUserMaster BOD is not published.

Configuring user access to your product through Infor Ming.le

You must complete this task before users can access your product through Infor Ming.le.

- 1 In Infor Ming.le, select **User Menu > User Management**.
- 2 Select **Security Administration > General Settings**.
- 3 Configure these general properties to ensure that new or changed user information is synchronized with other applications:

Publish Security User Master BOD

Select this option to publish the SecurityUserMaster BOD when a user's details are changed in Infor Ming.le. This synchronizes the user details with other products that subscribe to the BOD.

User Synchronization Interval (min)

If you selected the **Publish Security User Master BOD** option, specify the interval, in minutes, at which you want to publish the updated BODs to synchronize users.

- 4 Set up system administrator access between the Infor Ming.le portal and Mongoose. See [Configuring system administrator access to your product through Infor Ming.le](#) on page 31.
- 5 Add users in the Infor Ming.le portal:
 - Add users and assign them to roles through Infor OS User Management. See the *Infor Operating Service Administration Guide*.
 - If your product already has legacy users set up, you can import the users into Infor OS from a CSV formatted file. See the *Infor Operating Service Administration Guide*.
To temporarily disable user synchronization before the import, clear the **Publish Security User Master BOD** option and select **Do not enable** for the **Enable SCIM Parameter configuration** option. Then select **Allow Import**. The **Import** option is then enabled on the **Manage > Users** page.
- 6 Set up automatic login.
See [Configuring automatic login](#) on page 20.
- 7 Log in to Mongoose and update the user information in your product to specify additional required or optional fields that are not shared with Infor OS.
For example, on the **Users** form you must assign the appropriate license modules to the user, and you can assign the user's form editing permission level, PO requisition limit, or multi-site group authorization.
- 8 In Infor Ming.le, specify the Person ID that will be used to synchronize personnel records with Mongoose.
- 9 Optionally, Mongoose can send a Sync Person BOD back to Infor Ming.le to map the business user in Mongoose to a security user in Infor Ming.le.

Configuring system administrator access to your product through Infor Ming.le

Infor Ming.le provides a default Infor-SystemAdministrator role that allows for full access to all products that are linked to Infor Ming.le. Mongoose has an Infor-SystemAdministrator group or role that allows access to all Mongoose screens. When a user who has the Infor-SystemAdministrator role in Infor Ming.le is synchronized to Mongoose, that user automatically has access to all Mongoose forms.

- 1 Send the Sync SecurityRoleMaster BOD from Mongoose to synchronize the Infor-SystemAdministrator group or role in your product with the Infor-SystemAdministrator role in Infor Ming.le.
- 2 Add the Infor-SystemAdministrator role to the appropriate users in Infor Ming.le.
After you update the user information, the Sync SecurityUserMaster BOD is sent from Infor Ming.le to your product to synchronize the system administrator users with this role.
- 3 Assign additional system administration roles that are required by your product to the appropriate users.

Setting up roles or groups

Mongoose has configurable security roles that are set up in the product and then sent to Infor Ming.le through the SecurityRoleMaster BOD. This BOD is defined at the tenant level.

- 1 In Mongoose, use the **Groups** form to create roles that can access specific forms, programs, and logical folders.
A Sync SecurityRoleMaster BOD is generated by Mongoose. The information from this BOD is used in Infor Ming.le to automatically create a matching security role.
- 2 Associate the roles with users. This can be done in either Infor Ming.le or Mongoose:
 - To associate roles with users in Infor Ming.le, see the *Infor Operating Service Administration Guide*. Infor Ming.le generates a Sync SecurityUserMaster BOD and sends it to all applications that run in the Infor Ming.le portal. Mongoose uses the information from this BOD to associate the role or group with the Mongoose user record. This is the recommended method. Each application in the tenant should subscribe to the Sync SecurityUserMaster BOD.
 - To associate roles with users in Mongoose, use the **Users** form to associate each user with the appropriate groups. Mongoose generates a Process SecurityUserMaster BOD which is sent to Infor Ming.le. Infor Ming.le then sends a Sync SecurityUserMaster to all applications that run in the Infor Ming.le portal, to update the user records in all applications.

Note: If you manually create a security role in Infor Ming.le that does not exist in Mongoose, you must manually create a matching role in Mongoose.

If you delete a role in either product, you must also delete the corresponding role in the other product.

Person IDs

Any Mongoose users who plan to use Infor Ming.le or other Infor Operating Service components must be set up as users in Infor Ming.le, and the user's Person ID must be synchronized between Infor Ming.le and Mongoose.

Person IDs in Infor Ming.le

The user details page in Infor Ming.le includes an **ERP Person ID** tab. The **ERP Person ID** tab is populated if Mongoose publishes the Sync Person BOD. The ERP Person ID is used as the identifier of the user in the ERP application, and allows you to link the Infor Ming.le user to the ERP application user. The **ERP Person ID** tab shows all ERP person IDs that are associated with this Infor Ming.le user.

The IFS Person ID is used as the user identifier in Infor ION, which then links the Infor Ming.le user with the Infor ION BODs. By default, the IFS Person ID is set to the User Principal Name (UPN) of the user. UPN is an AD FS concept.

The IFS User Globally Unique Identifier (GUID) is the same as the UPN.

Sending the Person ID from your product to Infor Ming.le

When any of this information is manually added or updated in Mongoose for a user, a BOD is published by Mongoose.

The **Workstation/Domain ID** field on the **Users** form in Mongoose contains the Person ID for a user. It is possible to update this value in Mongoose and then manually publish the Process SecurityUserMaster BOD to infor Ming.le; however, Infor Ming.le is the system of record for user information, so you should only maintain the Person ID value in Infor Ming.le

Setting up distribution groups

Set up distribution groups in Infor Ming.le for Mongoose users who require access to ION in order to view or update information or to troubleshoot errors. See the *Infor Operating Service Administration Guide*.

Service accounts

The Infor Ming.le Service Accounts page shows accounts that have been created to allow applications a resource owner grant to contact the Infor Authorization Service to obtain a token for use in making API requests.

Chapter 8: Configuring your product and ION to send and receive BODs

You must configure your product to send and receive BODs.

Configuring your product for ION

You must set up this product so that it can generate BODs and place them in a message outbox where Infor ION Connect can retrieve them.

You must also set up the inbound message configuration, so that BODs sent to this application can process those messages.

Verifying that the SQL Server no count option is not set

In order to send all of the BODs from Mongoose, the **no count** option must be cleared in SQL Server Management Studio:

- 1 In SQL Server Management Studio, right-click on the Mongoose database server name and select **Properties**.
- 2 Select the Connections page.
- 3 In the Default Connection Options section, clear the **no count** option.
- 4 Click **OK**.

Setting up a default configuration for each site

On the **Sites/Entities** form, specify a default **Configuration Name** for the site. You must create a configuration with this name, which will be mapped to this site record.

Alternatively, if you leave this field blank, you must create a configuration whose name exactly matches the local site name, including the capitalization. For example, if your local site name is "oh", you must

create a configuration named "oh". You can use the Copy feature of the Configuration Manager utility to create this configuration.

Stop and start all Infor services after you create the configuration.

Setting up BOD replication and logical IDs, tenants, accounting entities, and locations in Mongoose

Use these steps to configure information that is required for BOD replication in Mongoose.

Note: After you perform these steps on one site, some information is replicated to the other linked sites.

- 1 If multiple sites on a Mongoose intranet will send or receive BODs, make sure that replication is set up between the sites as specified in the *Infor Mongoose Multi-Site Implementation Guide*. The sites must be replicating the Site Admin category to each other. On the Sites form at each site that will send or receive BODs, verify that all of the other sites are linked to this site.
- 2 On the Intranets form, to create a new intranet for Infor ION (bus), specify these values:

Intranet

Specify the name to identify the intranet, for example, **InforBUS**.

Description

Specify an appropriate description, for example, **Infor ION connection**.

External

Select this check box.

Transport

In this field on the **General** tab, specify **ESB**.

- 3 On the Intranets form, select the intranet used by your Mongoose site and specify this information:

Transport

Specify **HTTP**.

Queue Server

Optionally, specify the name of your MSMQ Server. You only need to specify a special queue name here if your site names are not unique, or if the MSMQ is on a different machine than the utility server. The queue server name can encompass both the name of the machine and the prefixed names of the queues.

If this field is left blank, the default queue server names are used.

Tenant ID

To use the same tenant ID for all sites on this intranet, specify it here. You can also leave it blank here and specify the tenant ID on the local site records, or leave that field blank to use the default tenant ID value of **infor**. The tenant ID can be a maximum of 22 characters and must match the tenant ID used in ION for other connecting products. For more information about how the tenant ID is used with ION, see the *Infor Operating Service Administration Guide*.

- 4 On the Mongoose application server, specify configuration information on the **Replication** tab of the Service Configuration Manager utility:
 - In the **Replicator/Inbound Bus Configurations** field, specify all of the configurations that can receive inbound or outbound messages from other sites on the intranet. The Replication Document Outbox form at each site collects outbound BODs for that site. The Replication Document Inbox form at each site collects inbound BODs for that site.
 - In the **ReplQListener Queue Servers** field, set up a queue server name for every intranet in your system.

Generally, the queue server name is set to `localhost\private$` which uses the standard private message queues named inbound and outbound on the utility server. However, if you have specified a **Queue Server Name** on the Intranets form, that name is used to build a path. For example, `utilityserver3\private$\MyQueues` results in the use of the queue names `MyQueues_inbound` and `MyQueues_outbound`.

Note: If you are upgrading from a previous version and you already have replication set up through a “bootstrap” site, it will still work as it did in the previous version. The **Replicator/Inbound Bus Configurations** field contains one site set as the Master Site, and the **ReplQListener Queue Servers** field has the **Default** field selected.

- 5 On the Sites form, create a logical “site” for Infor ION:

Site

Specify a name to identify this site as your ION site, for example: **InforBUS**.

Site Name and Description

Specify an appropriate site name and description.

Type

Specify **Site**.

Intranet Name

Select the intranet you created in step 3.

From Site

On the **Site User Map** tab, specify the local site, for example, **oh**. The site name is used as the accounting entity in BODs.

User Name

Specify the user ID used to send replication documents to the Mongoose outbox for ION to retrieve. This user must already be set up on the Mongoose Users form at the local site. We recommend that you specify the `repl_user` here, if it is defined. Otherwise, specify a user with Full User editing permissions. Also, the Automation (MGCoreAutomation) license module must be assigned to the user that you specify here.

- 6 On the Sites form, configure the local site, for example, **oh**, to generate or receive business object documents (BODs).
 - a Specify the **Message Bus Logical ID** for the site. This identifies the Mongoose site to ION. Specify it as `tenant.mongoose.site` where `tenant` is the tenant ID, `mongoose` is the application type, and `site` is the name of the local site, for example `infor.mongoose.oh`. The logical ID must be all lowercase.

Note: The logical ID entered here must match the application type and name defined in ION Connect for this Mongoose site.

- b Optionally, specify a **Tenant ID** for the local site. If no tenant ID is specified here or on the Intranets form, the default tenant ID value is **infor**.
- 7 At each site that is sending or receiving BODs, open the **Replication Rules** form and create a new rule. Specify this information:
Source Site
Specify the local site that you configured in Step 6.
Target Site
Specify the ION site you created in Step 5.
Category
Specify **ESB**.
Interval Type
Specify any option except **Transactional**.
- 8 At each site that is sending or receiving BODs, open the Replication Management form and click **Regenerate Replication Triggers**.
Note: If you have multiple sites in one database, you only need to perform this step at one site.

Setting up information that will flow to the other product

The values that you configure in Mongoose - for example, user records, or system parameters – often depend on the product with which Mongoose will exchange information. For more information, see the appropriate product integration guide.

Verifying BOD replication

To verify that replication is set up properly:

- 1 On the Mongoose application server, stop and start the Replication and Replication Queue Listener services.
- 2 In Mongoose, open a form whose information is being replicated to ION.
For example, the Vendors form information might be replicated.
- 3 Change a value and save the record.
For example, you could change the vendor address.
- 4 Check the Replication Document Outbox form to see if a BOD was generated and marked as processed.
In our example, the SyncSupplierPartyMaster BOD is generated.

After Mongoose is connected to ION, the BODs might be removed immediately from the outbox after ION retrieves them. In that case, you can check ION OneView to verify whether the BOD was sent and processed.

Connecting your product to ION

A connection point must be set up for Mongoose so that it can send documents to ION.

Optionally, you can then set up and activate a document flow between two product connection points.

Understanding connection points and document flows

You create connection points in ION Desk. Connection points provide the information that ION uses to connect to a product's message inbox and outbox. At least one connection point must be defined for each product instance that integrates to ION.

For each connection point, you select the BOD documents that can be sent or received by the product instance. These correspond to the BODs that are listed in [BODs used in integrations with this product](#) on page 83.

In an integration between two BOD-enabled products, document flows are set up to define the BODs that flow between the application connection points.

A connection point can be reused multiple times in one or more document flows. You can also create connection points during the modeling of a document flow in the details section of the document flow elements. The connection points that are created during the modeling of a flow are added to the shared list of connection points, and they can be reused.

Some integrations with Mongoose use a solution XML file to set up sample connection points and document flows. This file can be imported to ION to provide the basis for connection points and document flows.

See the section on ION Connect Modeling in the *Infor ION Desk User Guide* for additional information about connection points and document flows.

Creating a connection point

- 1 In ION Desk, add a connection point for each instance of the product.

See the *ION Desk User Guide*.

Use this information to complete these fields:

Name

Specify the Mongoose site ID.

Type

Specify **Infor Application**.

Logical ID Type

Specify **mongoose**.

Tenant

Specify a tenant, or leave blank to use the default tenant infor.

Database Driver

Specify **SQL Server (Microsoft)**.

- 2 Specify the Mongoose application database connection details, including the database server, port, schema, and connection credentials.
Other advanced settings are available.
- 3 Select **Delete Processed Messages** to immediately delete BODs from the Mongoose Replication Document Outbox after ION processes them.
Otherwise, the processed BODs remain in the outbox for the number of minutes specified in the **Outbox Cleaner Expire Time** field.
Cloud difference: This option is still valid for ION 12, both for on-premises and for ION CE, as long as the connecting application is on-premises. But if both ION and the connecting application are in the cloud, then this option is not shown in ION.
- 4 Expand the **Advanced** section at the bottom of the pane.
- 5 Under **Message Processing Preference in Outbox/Inbox**, select both **By Tenant** and **By Logical ID**.
- 6 Test and save the connection point.

Connection points are not activated separately. When you activate a document flow, the associated connection points are activated.

Defining product documents for the connection point

- 1 In ION Desk, select a Mongoose connection point.
- 2 Click the **Documents** tab of your Mongoose connection point and select all of the documents that can be sent or received by this product instance for any integration. For each document, specify whether it can be sent, received, or both.
See the list of inbound and outbound documents in [BODs used in integrations with this product](#) on page 83. The list of documents that you define for this connection point must match that list.
You can also add custom document types.
See the *ION Desk User Guide*.

Note: If you later remove a document that is used in a document flow, an error is reported in the document flow messages pane.

Configuring a document flow between products

This section provides general steps for setting up a document flow in ION Desk. See the *Infor ION Desk User Guide* for detailed steps.

Note: Usually, document flows are configured and activated in conjunction with integrations between this product and another product; in that case, skip this section and follow the steps in the appropriate integration guide. If you want to send BODs to another ION-enabled product but there is not an appropriate integration guide, follow the steps in this section.

- 1 Define a new document flow that includes connection points for both Mongoose and the other application.
- 2 Define the documents that flow between the applications, and the direction that they flow.
- 3 Save the document flow.
- 4 Activate the document flow.

Message listeners

As an alternative to setting up document flows to products that are supposed to receive every published BOD, such as Infor Ming.le and Infor Business Vault, message listeners can be activated to automatically forward all BODs for specified verbs to these products.

See the *Infor ION Desk User Guide* for information about message listeners.

Publishing BODs

After you define the Mongoose connection point in ION, you publish the associated BODs so that ION can retrieve them from your message outbox. You must publish BODs for general use by any product that subscribes to the BODs through Infor ION.

For an ION-based integration between this product and another product, see the appropriate integration guide to set up the document flows and to perform any additional configuration of the products that is required for that integration. You can then use the steps in the integration guide to publish the BODs.

Starting products and services

Start these products, services and processes if they are not already running:

- Mongoose
- Other products with which you are communicating through Infor ION
- ION Service

Use the ION Grid Management user interface to verify that the ION Service is running. See the *Infor ION Grid Administration Guide*.

Sending the initial data load

During normal processing, BODs are generated when a user makes a data change or completes a transaction. However, when you start a new integration with another product, you must manually generate BODs that pass the current set of data from your product to another product, so that the base systems are synchronized. The BODs are placed in the message outbox, from which Infor ION retrieves them and passes them to the other product.

To perform an initial data load of into BODs that are placed in the message outbox:

- 1 Open the Replication Document Manual Request Utility.
- 2 Select **Initial Load** so that the data is processed as an insert, that is, as new data.
- 3 Select the BODs to include in the data load.
The BODs you select are different for each integration. If you select multiple process BODs at one time, the BODs are processed in alphabetical order. However, you must load these BODs in the order specified in "BOD dependencies" to prevent errors in the integration due to missing data.
- 4 Click **Process** to process the selected BODs.
- 5 Continue to select and process BODs until all of the BODs are processed that are required for this integration.
- 6 Repeat this process at each Mongoose site.

When you use this utility to publish BOD data, the current date is used for all Status/EffectiveDateTime elements.

After the initial data load, Mongoose sends BOD updates when users change certain data or when certain transactions occur in Mongoose.

For a list of the events that generate BODs, see [Business events that generate outbound BODs](#) on page 71.

Verifying the ION configuration

Verify that the connections are set up correctly.

Verifying that BODs are generated

- 1 Use the information in [Business events that generate outbound BODs](#) on page 71 to determine what user actions generate certain BODs.
- 2 Perform those user actions in Mongoose.
- 3 In ION Desk, review the messages by using ION Desk OneView or other ION Desk functions.
- 4 If messages are not displayed in ION Desk, check the the Mongoose Replication Document Outbox form to see if the BODs were generated but not retrieved by ION.
- 5 If messages are not flowing, review the XML data for the BOD.

Verifying that ION receives data

To verify that ION is receiving BODs from Mongoose, monitor these areas in Infor ION Desk:

- Select **Connect > Active Connection Points** to show the message queue counts.
- Select **OneView** to show individual messages, if they are enabled.
- Select **Error BODs** to see if any errors were reported.

If you encounter problems, see [Data is not flowing properly](#) on page 69.

Verifying the data flow between products

After you create and activate document flows to other products, you can check the message inbox area of the other products, to ensure that data is flowing between the systems.

Chapter 9: Configuring drillbacks to your product

You must configure drillbacks to your product.

Importing and configuring application drillbacks

To allow Infor Ming.le users to open applicable screens in Mongoose directly from an Infor Ming.le object, you must import an XML document to create the mapping from the object type to the appropriate Mongoose form.

Note: The document flow between Mongoose and Infor Ming.le must be activated in Infor ION so that drillbacks can work.

Available drillbacks

“Drillback” refers to the ability to launch a Mongoose page in Infor Ming.le from a link based on the document ID for the relevant BOD. A link is defined by a view for each type of BOD noun and is then configured to launch a specific Mongoose page for that noun. Not all nouns have a drillback view available.

Loading the BODs for drillbacks

In order to use these drillbacks, you must perform an initial load of the BODs listed in [Available drillbacks](#) on page 43, using the Replication Document Manual Request Utility in Mongoose. For more information, see [Sending the initial data load](#) on page 41.

Chapter 10: Configuring context apps and utility apps that are used with your product

Context apps are lightweight applications that communicate with the application frame to publish contextual information to the user. These applications subscribe to information published by the application frame and display relevant content only when it is available. The information in the applications is context-sensitive, which means that the values or content in the application depend on the current context (form or field) in Mongoose.

Utility apps are lightweight applications that represent information unrelated to content in the application. They do not communicate with the application frame and, if activated, show only when the application is open.

Understanding business context messages

A business context message is a JSON message that is sent from a Mongoose screen, field or other user interface object for the purpose of establishing the current context. These messages can be used for many contextual purposes within Infor Ming.le, as well as for drillbacks.

Mongoose uses business context messages to display online help and to determine the metrics and reports to display in the In-Context BI application when a certain Mongoose screen is displayed.

To find the business context message that is sent by a specific object, use the Infor Ming.le Context Viewer application. See the *Infor Ming.le User Guide*.

Process default settings that allow context messages to be sent

In Mongoose, these options are set on the **Process Defaults** form:

- **Send Context Message on Form Focus** enables sending of business context messages for forms. Default is 0 (not sent), but it is automatically set to 1 for Mongoose through initialized data. When set to 1, the context message "screenId": "mongoose_Formname" is sent for all forms, where *Formname* is the form name, for example, "screenId": "mongoose_UsersMaint".
Note: This process default is set to 1 so that Mongoose context apps work in Infor Ming.le.

- **Optional Context Message Form Prefix** replaces the Mongoose default prefix on all forms with a specified prefix. For Mongoose, the prefix is set to `mongoose_`. So, for example, the `UsersMaint` form is sent as `mongoose_UsersMaint`.

Supported business context messages

Business context messages contain sets of contextual information called entities, which include information similar to that of Infor BODs. Mongoose can pass information about multiple entities in one business context message. This example shows an `inforBusinessContext` context message that was sent from the `Customers` form in a Mongoose-based application:

```
{ "type": "inforBusinessContext",
  "data": { "screenId": "CSI_Customers",
    "entities": [
      { "entityType": "InforAccountingEntity", "id1": "DALS" },
      { "entityType": "InforShipToPartyMaster", "id1": "4851", "id2": "0", "accountingEntity": "DALS", "drillbackURL": "?LogicalId=lid://infor.syte line.dals&page=formonly&form=Customers (FILTER (Cust Num%3D%27%20%20%20%20%2013%27%20AND%20CustSeq%3D0) SETVARVALUES (InitialCommand%3DRefresh))", "bodReference": { "noun": "BillToPartyMaster", "documentId": "4851", "accountingEntity": "DALS", "logicalId": "lid://infor.syte line.dals" } }, { "entityType": "InforCustomerPartyMaster", "id1": "4851", "name": "Brand Central Dept Store", "accountingEntity": "DALS" } ] }
```

The message informs context apps that the `Customers` form is currently displaying a record for the `Brand Central Dept Store`, which has the customer ID `4851` and the `Ship To` address ID `0`, in the accounting entity `DALS`. The message contains one or more business entities, or collections of information, associated with the current record. The entity information can include a relative URL that can be used to drill back into the application and display the form and record. It also can include a BOD reference that maps from the current entity in the user interface to a corresponding BOD noun.

Any context apps that are associated with Mongoose receive all of the information in this message. Each context app searches its data to determine whether it has contextual information to display that is related to information in this message.

Supported context and utility applications that are delivered with Infor Ming.le

The `Manage Context/Utility Apps` page lists all of the default context and utility apps and other context and utility apps that you have added. From this page, you can add and view context and utility app details.

These context and utility applications are available by default for use with Mongoose:

- Alerts utility application and homepages widget
- Context Viewer context application
- Documentation context application
- In-Context Business Intelligence
- Paparazzi context application
- Posts utility application
- Related Information context application
- Share from the Infor Ming.le shell
- Smart Help context application
- Social objects
- Tasks utility application and homepages widget

To know how to use the standard applications, see the *Infor Ming.le User Guide*.

To know how to use the Related Information context application, see the *Infor Document Management User Guide*.

To know how to set up and use the Smart Help context application, see the *Infor Smart Help Administration Guide* or online help.

Note: The Smart Help context application displays the help topic associated with the currently displayed form. You can open the Mongoose online help from a link.

Depending on the options you select in the Smart Help application and your permissions, you can view links to Knowledge Base articles or training materials related to the form displayed. You can also view or add links to the customized help that your company has created for the form in Infor Document Management. See *Infor Smart Help Administration Guide*.

Adding a context or utility app to your product

You can add a context or utility app to your product if it is not set up as a default. See the information about adding context apps or utility apps in the *Infor Platform Xi Administration Guide*.

Either select **Grant access to all users** or click **Add new users and/or IFS security roles** to specify certain users or roles in Mongoose that can access this context app.

Context applications that are delivered with your product

The In-Context KPI context application is included when you add the Mongoose application in Infor Ming.le. This context application is available in addition to the context applications that are delivered with Infor Ming.le.

Configuring the In-Context KPI app

To enable the In-Context KPI app for the Mongoose application:

- 1 In Infor Ming.le, edit the InContext KPI context app.
- 2 In the **URL** field, specify
`https://mg_application_server/WSWebClient/WSChildForm.aspx?notitle=1&Form=DisplayCriticalNumber`
Replace `mg_application_server` with the Mongoose application server name.
- 3 Click **OK**.
- 4 Grant access to all users and ensure that this app is enabled for the Mongoose application.
- 5 Authorize users to display the Critical Numbers.

Authorizing users to display critical numbers in widgets or the InContext KPI app

To enable users to view and select critical numbers for display in widgets or home pages, you must set up the appropriate authorizations:

- 1 In Mongoose, open the **Critical Number Users** form.
- 2 Select a user record.
- 3 On the **Categories** tab, specify user authorizations for a critical number category, or on the **Permissions** tab, specify user authorizations for specific critical numbers.
- 4 Save your changes.
- 5 Ask the user to log out of Mongoose and then log in again, so the changes can take effect.

How critical numbers are displayed in the In-Context KPI app

When the user opens a Mongoose form in Infor OS, the form sends a JSON `inforBusinessContext` message that includes its entity type. The entity type is the BOD sent when the content of the form is update.

These actions are taken when the business context message is received:

- Check the entity type in the business context message and find the related category of forms.
- Determine which critical numbers (if any) the user has selected within that category.
- Determine how many instances of the In-Context KPI app the user has created in Infor OS.
- Load the selected critical numbers into multiple instances of the Display Critical Number form. Each instance is displayed in an In-Context KPI pane.

The sequence of the critical number determines which critical numbers are shown. For example, if the user selected five critical numbers for a category, but only added three instances of the In-Context KPI app, then only the first three (sequential) critical numbers are displayed.

Chapter 11: Configuring Homepages for your product

This section describes administrative tasks that are required to set up Homepages with Mongoose in Infor OS.

For general Homepages administration tasks, see the *Infor Operating Service Administration Guide*.

Setting a default theme so that widgets display properly

In the Mongoose Configuration Manager utility, set a default theme for all users. Users can change the theme later, using the User Preferences form in Mongoose. The theme value must not be blank; otherwise, problems occur with the display of widgets on the homepages.

Chapter 12: Configuring workflows and ION messages for your product

You can configure workflows and ION messages for your product. Example workflows are provided.

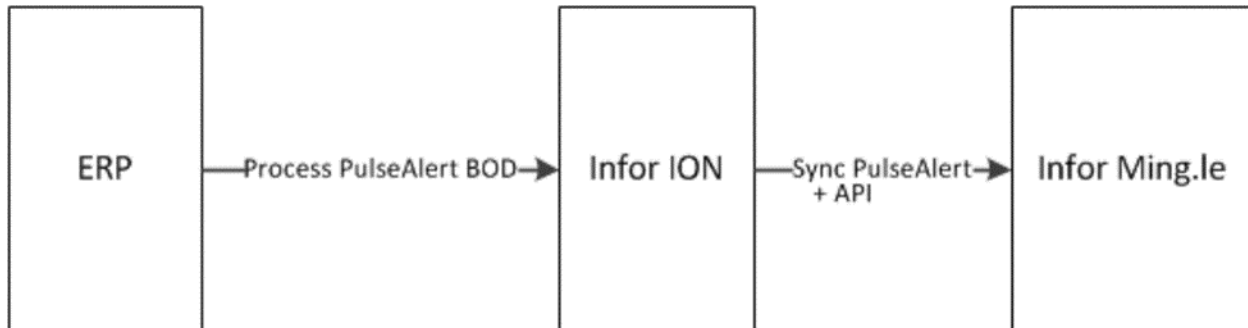
ION messages

This table shows the types of ION messages:

Type	Description
Alert	In Infor ION Event Management, you can monitor business events and create alerts when exceptions occur. BODs are published by applications when business events occur. Monitors are processes that run in the Event Management engine and evaluate these BODs by applying predefined business rules: comparison conditions, value change conditions, and document overdue timer based rules. When exceptions are detected, these are reported as alerts to the business users. The distribution list of alerts is included in the monitor definition. Users can receive alerts on their homepages or by email.
Task	A task is a workflow step that creates an entry in a user's task list. The workflow stops until the user has completed this task, and then the next step in the workflow begins.
Notification	A notification is a workflow step that creates a notification in the user's task list. The workflow continues after the notification is sent to a user. You can configure notifications to perform these actions: • Send a message to inform the user that a certain point in the workflow has been reached. • Optionally, include notes from previous steps. • Display workflow parameters and structures. A distribution list can be defined for notifications.

About alerts sent from this product to Infor Ming.le

Mongoose can send PulseAlert BODs to Infor ION that can be displayed as alerts in Infor Ming.le, as shown in this process flow:



A Process PulseAlert BOD is generated when a business event associated with a publication occurs in Infor Mongoose.

Infor Mongoose users can subscribe to the appropriate publications. Then, when an alert occurs and a BOD is generated, the email addresses for all subscribers are included in the BOD.

The email address that is used in the BOD is determined by the **Email Type Description** field on the Users form. If an address is specified where the Email Type Description is set to **PersonID**, then the email address associated with the PersonID type is used. If no PersonID type is found, then the address in the **Primary email Address** field is used, as shown in this screen:

Groups	Additional Info	Login Information	E-mail Address	Source Control
Primary E-mail Address: Work bob.smith@infor.com				
☐ Use Reply To 				
☐ Send External Notifications Work bob.smith@infor.com				
☐ Send External Prompts Work bob.smith@infor.com				

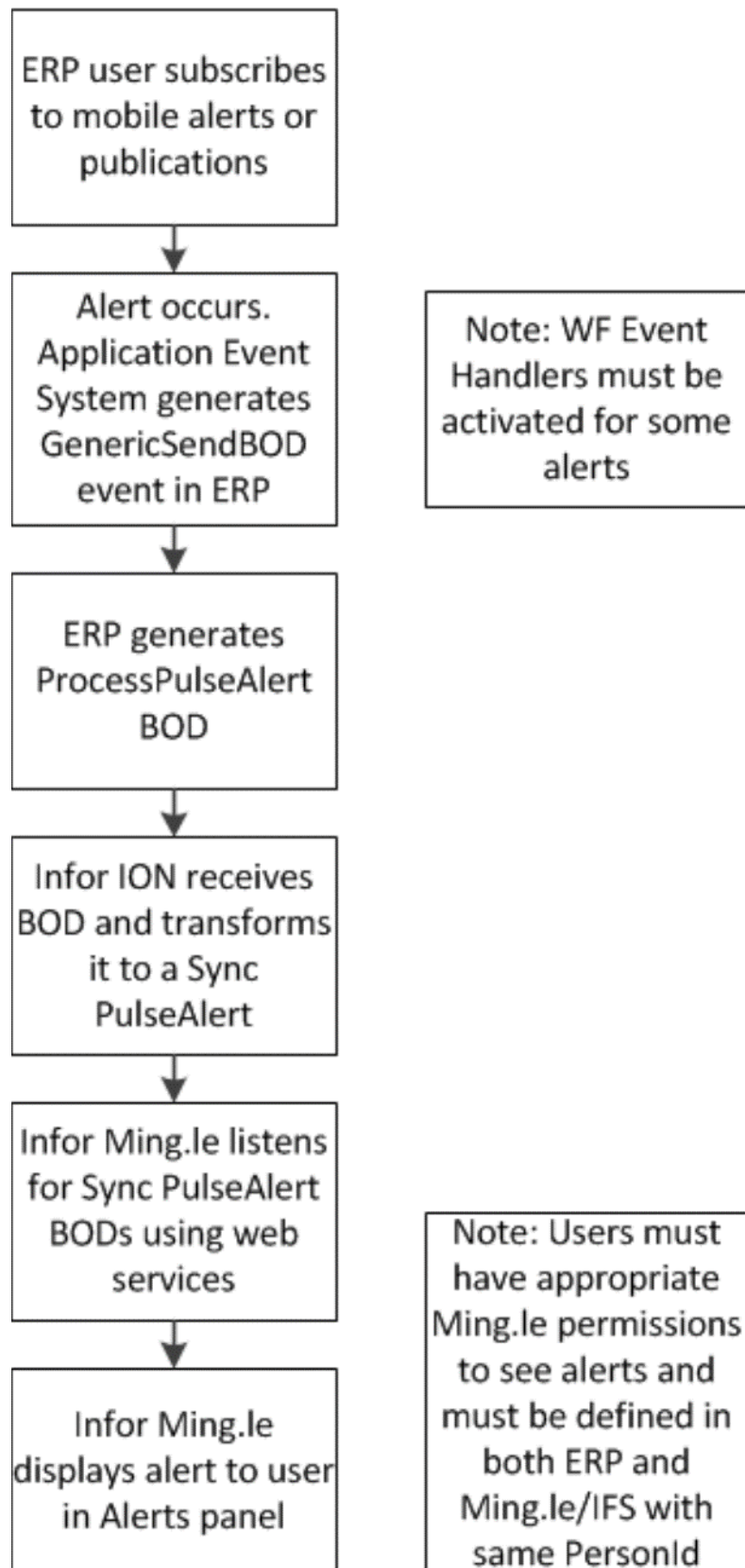
	* E-mail Type Description	* E-mail Type Address
1 ▶	PersonID	bsmith@infor.com
2	Work	bob.smith@infor.com

If neither of these areas are populated, then a BOD is not sent.

Infor ION attempts to match the email address from the BOD with a PersonID or primary email address that is defined for a user in Infor Ming.le, so that it can pass along the alert. If a matching address is

found, and if the user has the appropriate Infor Ming.le permissions, the alert is displayed on the user's Alerts panel in Infor Ming.le.

This is the general process flow:



Configuring monitors for alerts

1 Configure this product to send alerts.

- a For publications that have associated event handler sequences in the table above, activate the sequences so that the BOD will be sent when the events fire. In the **Workflow Event Handler Activation** form, select the appropriate event and sequence, and select **Active for this Site**.
- b Mongoose users must subscribe to the appropriate publications from the list in the table above, through either the **Mobile Alerts** form or the **Publication Subscribers** form.
- c Set up Mongoose users with either a PersonID email address or a primary email address, or both. In the **Email Addresses** tab on the Users form, add a record to the grid where **Email Type Description** is set to **PersonID** and an email address is assigned. You can optionally set the **Primary Email Address** to another email address for the user.
- d See [Person IDs](#) on page 32.

2 Configure ION to process alerts from this product.

You cannot create connection points for Infor ION Pulse, so there is no document flow that includes Pulse. If an application connection point that can publish ProcessPulseAlert is used in at least one active document flow in ION Desk, the BOD is delivered directly to ION Pulse.

In order to send alerts from your product, you must add the ProcessPulseAlert BOD to the Document tab of your Mongoose connection point as a document that can be sent from the site. Through an internal API, ION passes the alert to Infor Ming.le as a Sync.PulseAlert BOD.

See [Defining product documents for the connection point](#) on page 39.

See the *Infor ION Desk User Guide*.

3 Configure Infor Ming.le to display alerts for this product.

Set up Infor Ming.le users with the appropriate roles to access tasks and alerts. See information about configuring user access for tasks and alerts in the *Infor ION Desk User Guide*.

In addition, the value in the Infor Ming.le and IFS Person field must match the **PersonId email address** or **Primary Email Address** for the same user in Mongoose. This is handled automatically if you set up users first in Infor Ming.le/IFS and then use SecurityUserMaster BODs to populate the user records in Mongoose, as described in [Users and roles](#) on page 27.

Configuring workflows for tasks and notifications

1 Configure this product to send tasks and notifications.

Use the sample workflows described later in this chapter to understand this process.

2 Configure ION to process tasks and notifications from this product.

See the *Infor ION Desk User Guide*.

3 Configure Infor Ming.le to display tasks and notifications for this product.

See the *Infor ION Desk User Guide*.

Workflow example content for your product

This section provides information about the event monitors and workflow models that were created specifically for BODs published by Mongoose. It is intended mainly for system administrators who manage and customize standard ERP ION content, or who must troubleshoot or trace back tasks or alerts to the ERP.

Recommendations are given on how to customize the standard ION content to suit your business needs. We recommend that you discuss the process and results of your planning decisions with Infor Consulting Services prior to implementation.

If you plan to create custom tasks, workflows or alerts that are generated from your product to run in Infor ION and Infor Ming.le, see the *Infor ION Development Guide*.

Setup and configuration

This section describes how to install the Mongoose ION workflow example content. It also describes the customization process.

Requirements

To use the Mongoose ION workflow example content, at least one document flow must be activated for the Mongoose connection point. The solution XMLs that are provided for specific integrations of other products with Mongoose contain a document flow with a single application for the Mongoose connection point. You can select multiple connection points in the document flow, if required. If a monitor or activation policy is activated, subscriptions are created for each active Mongoose connection point. See the *Infor ION Desk User Guide* for more information.

Installing the content

Use this procedure to install the ION distribution groups, workflows, monitors, and activation policies:

- 1 Download the SL_ION_Content.zip file from the Infor Xtreme Support portal.
You can find this file attached to solution 1440389.
- 2 Extract the files.
- 3 Log into Infor Ming.le as a user with the DataAdministrator role.
- 4 Ensure that the users and roles that will use the workflows and alerts exist in Infor Ming.le and Mongoose.
See [Configuring user access to your product through Infor Ming.le](#) on page 27 and [Setting up roles or groups](#) on page 31.
- 5 Import the distribution groups from the SL-DistributionGroups.csv file.
- 6 Assign users to the appropriate distribution groups.

See the *Infor Operating Service Administration Guide*.

7 In ION Desk, import these files and activate the monitors, activation policies, and workflows:

- SL-Monitors.xml
- SL-ActivationPolicies.xml
- SL-Workflows.xml

See the *Infor ION Desk User Guide*.

Configuring drillback views for Workflow modeler

Configure drillback views. See the *Infor ION Desk User Guide*.

Distribution of alerts and tasks

ION content includes monitors, activation policies and workflows. Monitors create alerts and workflow models create tasks. Alerts and tasks can be distributed to named users, distribution groups, or to persons defined in BODs. The standard Mongoose ION monitors and workflow models use distribution groups and persons defined in BODs.

The value in the email address field of the Infor Ming.le user is used when emails are sent.

Updating Mongoose content for ION

Use ION Desk to copy and adjust standard content. Updates do not overwrite imported content. When you import a monitor, workflow or activation policy that already exists, you can either select Skip or Copy:

- If you select **Skip**, the import does not take place.
- If you select **Copy**, the import takes place but the imported object is renamed and the existing object is preserved.

To replace content, delete the standard content before importing a new Mongoose content pack.

Chapter 13: Configuring your product to work with Infor Document Management

Infor Document Management and Mongoose are integrated through these methods:

- ION API (rest-enabled)
- Context Business Messages
- BOD nouns

The CaptureDocument BOD is an outbound BOD that is sent by IDM when capturing different types of documents, for example, scanning documents.

The ContentDocument BOD is both an outbound BOD and inbound BOD. Mongoose creates documents in IDM by sending a Process ContentDocument BOD. Mongoose receives updates to documents within IDM by receiving Sync ContentDocument BODs.

Mongoose delivers its own document models and set of links to the UI products (if applicable).

Both document models and links can be configured for a tenant.

Importing configurations and types

See the instructions in the *Infor Operating Service Administration Guide* to import a configuration using the Infor Document Management Control Center. You can import these objects:

- To import document types, use the Configuration Importer.
- To enable the Related Information context app to show Infor Document Management documents, import the Mongoose Business Context Model configurations. See the information about the business context model in the *Infor Operating Service Administration Guide*
- To import the ION configuration, use the ION option. The ION configuration enables communication between IDM and ION. You can also configure a specific ION workflow for a particular document type. See the *Infor Operating Service Administration Guide*

Enabling the Related Information context app for your product

When you set up the Related Information context app in Infor Ming.le, use the **Applications** tab to add the Mongoose application and set it to **Enabled**.

Chapter 14: Setting up ION APIs for your product

Using ION APIs to consume Mongoose REST services

The ION API service is a broker for Infor applications that use web service APIs. The service handles synchronous communications to application-specific APIs, versus asynchronous BOD communications that conform to the OAGIS standard.

Currently Mongoose does not require any configuration of ION APIs. However, you can configure any Mongoose-based application running in Infor Ming.le to allow calls to the Mongoose REST service through ION API.

Settings in the IDO Request Service `web.config` file determine whether the Mongoose REST service is enabled for use with ION API or is using the traditional Mongoose token exchange.

For more information about authenticating against ION API to consume Mongoose REST web services, see the *Infor Ming.le ION API User Guide*.

For more information on consuming Mongoose REST web services and which methods are available, see the *Mongoose REST Web Service* document or use the Open API documentation that is available in this area: https://mongoose_application_server/IDORequestService/MGRestService.svc/api-docs/.

You can set Mongoose REST as an API in ION API. You need to edit the `web.config` file for your application to store the credentials needed to access the Mongoose REST service.

Setting up Mongoose REST service as an API

To set up Mongoose REST service as an API in ION API:

- 1 Open the `web.config` file in the `IDORequestService` directory.
- 2 Modify these application settings:

`enableRestWorkstationLogon`

Specify `true` to enable this functionality. To disable, specify `false`.

`restOAuthConsumerKey`

Specify the consumer key value that is provided to ION API during provisioning (creating an entry in ION API)

`restOAuthConsumerSecret`

Specify the consumer secret value that is provided to ION API during provisioning

Using Mongoose to consume APIs from other applications through ION APIs

The Mongoose application user has an Infor Ming.le identity that is set up through Single Sign On (SSO). Mongoose can exchange the SSO credentials for an ION API token. This token is used to interact with other systems via ION API Gateway.

With the ION API token, Mongoose can invoke web services for other applications on behalf of that user, which are exposed through ION API. These include but are not limited to Infor Ming.le, Infor Document Management, and M3.

All interactions are performed with the user credentials in the source and target systems, complying with any security configuration, and ensuring that the audit trail is not compromised.

Chapter 15: Configuring Infor Data Lake for your application

Mongoose can send database table schemas, master data and transaction information to the Data Lake for use by other applications. For example, Birst takes information from the Data Lake to use in Analytics.

Mongoose uses the ION Messaging Service (IMS) to send JSON messages that contain master and transaction data to the Data Lake. Before you can use this functionality, you must perform some configuration in your application to create an intranet, a logical site, and a replication category that can be used to send the messages to ION and then onward to the Data Lake.

In ION Desk, you must also create an IMS connection point for the site that you configured in Mongoose. Then create a Data Lake flow that uses the connection point to send data from your application to the Data Lake.

Your application must also send its database table schema to the Data Lake, so that the Data Lake knows how to organize the master and transaction data that it receives. This is generally done when replication triggers are regenerated. But, if necessary, you can send the schema information manually.

About outbound messages

Based on generated trigger logic and the Replicator service, each time you add or update a record in one of the tables that is listed in the Data Lake replication category, data from the tables listed in the that replication category is sent to the Data Lake using JSON messages.

When you click **Generate Schema** on the **Generate Data Lake Schema** form, updated schema information for the tables listed in the DataLake replication category is sent directly to the Data Catalog URL.

Configuring your application for IMS and the Data Lake

Use these steps to configure your application to send data using ION Messaging Service (IMS) messages.

- 1 Use the **Intranets** form to add an intranet to use for IMS (and Data Lake) messaging. Provide this information:

Intranet

Specify a descriptive name, for example **IMS-DataLake**.

Description

Indicate that this intranet is used for IMS and Data Lake messaging.

External

Select this check box.

Transport

Select **ESB**.

ION Transfer Type

Select **ION Messaging**.

Data Catalog URL

Specify the URL for direct connection to the Data Lake Catalog (or other data repository), when sending table schema information directly to the Data Lake. For example:

`https://<server>/<tenant>/IONSERVICES/datacatalog`

You must also have this information available:

URL

Specify the URL to use IMS connections to ION. For example,

`https://<server>/<tenant>/IONSERVICES/api/ion/messaging/service`

For IMS messaging and Data Lake connections, this is the destination to which information is to be sent.

Tenant ID

Specify the tenant ID. This ID is used in all BODs and messages to identify the sender of the document. This ID is used primarily for transactions on hosted software or "Software as a Service" (SaaS). In a cloud environment, the tenant ID is automatically provided, and you cannot change it.

- 2 Use the **Sites** form to create a site to be used for IMS and Data Lake messaging.

- a Provide this basic information:

Site

Specify a concise name of eight letters or less, for example: **DATALAKE** or **IMS-DL**

This name is used internally by the system.

Site Name

Specify a more descriptive name for the site.

Description

Specify a description for the site, for example: **IMS-Data Lake virtual site**

- b On the **System Info** tab, provide this information:

Language

From the drop-down list, select the language to be used on the site.

System Type

Leave this field blank.

Time Zone

From the drop-down list, select **UTC**.

Intranet Name

From the drop-down list, select the intranet that you defined in Step 1.

Tenant ID

Provide the Tenant ID for your application, as specified on the **Sites** form.

- 3 Use the **Replication Categories** form to create a Data Lake category that holds the tables that to be replicated to the Data Lake.
 - a Add a **Category** named **DataLake**.
 - b From the **Replication Transfer Type** drop-down list, select **Data Lake**.
 - c In the **Table or Function** grid, specify the tables to include in the category.
 - d Save.
 - e Set the **Interval Type** to **Immediate**.
 - f On the **Replication Management** form, click **Regenerate Replication Triggers**.

Whenever a table listed in the replication category is updated, the replication functions send a message that contains the table update to the IMS URL that was defined for the IMS(Data Lake) intranet.

- 4 Use the **Replication Rules** form to define rules for Data Lake replication:

Provide this information:

Source Site

From the drop-down menu, select the site from which the data is to be sent.

Target Site

From the drop-down list, select the site you defined in Step 2, the destination to which the data is to be sent (typically, the Data Lake).

Category

From the drop-down list, select **DataLake**.

Description

Optionally, provide an internally used description such as **Data Lake**.

Interval Type

From the drop-down list, select **Immediate**.

Disable Replication

Leave cleared (not selected).

Update All Columns

Leave cleared (not selected).

Generating table schema for use in the Data Lake

Table schema are used to structure the master and transaction data that is sent through IMS to the Data Lake. Use this procedure to send initial or updated database table schema information to the Data Lake:

- 1 Open the **Generate Data Lake Schema** form.

Note: The fields on this form are all read-only. To change any of the values, you must go to wherever the values are set, as indicated in the next step.

- 2 Verify that the information in the fields is what you intend:

Site

This is the logical site that you defined on the **Sites** form for sending and receiving Data Lake (IMS) messages.

Intranet

This is the intranet that you defined on the **Intranets** form for sending and receiving Data Lake (IMS) messages.

Intranet Description

This is the value of the intranet description as provided on the **Intranets** form for the intranet that you specified.

Data Catalog URL

This is the value of the **Data Catalog URL** that is defined for the intranet that you specified.

URL

This is the value of the IMS **URL** that is defined for the intranet that you specified.

- 3 To send the table schema information listed in the Data Lake replication category to the **Data Lake URL**, click **Generate Schema**.

For example, if the replication category only lists 3 tables, then only the schema for those 3 tables are sent. If you modify any of these tables in Mongoose, you would need to regenerate replication triggers, and then click **Generate Schema** again to update the schema in the Data Lake.

Note: When you regenerate the replication triggers, the schema should be generated and sent automatically. However, should something go wrong or should you want to generate the schema manually, you can still use this button.

Configuring ION Desk

To send data to the Data Lake, you must establish a connection using the ION Desk and then create a data flow. Use the procedures in this section.

Verifying the schema as custom documents in the Data Catalog

To verify that the schema documents were sent to ION, you must be running the ION Desk as administrator. Then perform these steps:

- 1 From the **Data Catalog** menu, select **Custom Object Schema**.
For an initial tenant setup, it is likely that no documents are displayed on the page.
- 2 To populate the list, click the Refresh icon.
You should see schema documents of type JSON that match the table schemas that you generated in your application.

Creating the connection point

You must create, in ION Desk, a connection point for the IMS-DataLake site that you created for your application. The connection point must be of the type IMS, which authenticates messages using OAuth 1.0.

To create the connection point, in ION Desk:

- 1 Select **Connect > Connection Points**.
- 2 Click **Add** and select **IMS**.
- 3 Provide this information:

Name

Specify `mg_ims`.

We recommend using "ims" in the name, to distinguish from other, standard site connection points.

Location

URL

Specify the REST web service URL for your application, for example,
`https://<server>/idorequestservice/MGRestservice.svc`

OAuth Consumer Key (For URL)

Specify the OAuth consumer key provided by Infor.

OAuth Consumer Secret

Specify the OAuth consumer secret provided by Infor.

ION Endpoint Path

Specify the ION end point, for example, `https://<ION server>/api/ion/messaging/service`

OAuth Consumer Key (for Endpoint)

Specify the OAuth consumer key provided by Infor.

OAuth Consumer Secret

Specify the OAuth consumer secret provided by Infor.

- 4 Click **Generate**.
- 5 To verify that the connection is set up properly, click **Test Connection**.
This action calls the discovery Mongoose web method, which sends a list of the tables being exported from Mongoose into the Data Lake through IMS.
- 6 To add documents to the IMS connection point, click the plus (+) icon on the **Documents** tab.
- 7 From the **Type** drop-down list, select **JSON**.
- 8 Select all of the listed documents and click **OK**.

Creating the Data Lake flow

To create the Data Lake flow, in ION Desk:

- 1 Select **Connect > Document Flows**.
- 2 Click **Add** and select **Data Lake flow**.
Name the flow something like `mg_dataLake`.
- 3 Drag and drop an application onto the flow to the left of the Data Lake application and name it `mg_ims`.
- 4 In the **Application Activity Properties** panel, click the plus (+) icon and select the `mg_ims` connection point.
- 5 Click the document icon between the `mg_ims` and Data Lake applications.
- 6 On the **Documents** tab, add all of the available JSON documents.
- 7 Click **OK**.
- 8 Save your changes.
- 9 Activate the flow.

Verifying the Data Lake configuration

Use this procedure to verify that your configuration is set up correctly to send data to the Data Lake.

- 1 In your application, make a change to a form that affects a table listed in the Data Lake replication category.

For example, if the Data Lake replication category contains a User Master (user_mst) table, add a new user on the **Users** form.

This sends a JSON message to the replication tables. The message is then sent directly to ION and does not use the ReplQListener or the Replication Outbox. The message updates the information in the user table in the Data Lake.

- 2 Check to see if any replication errors were reported for the changed table.
- 3 Check the **ION Messaging Errors** form to see if any errors occurred during ION messaging. Problems could include an incorrect URL or a problem with the Replication Listener. If an error occurred, correct the error and select **Resubmit**. When you save the record, the message is resubmitted for reprocessing.
- 4 In ION Desk, check OneView to see whether the message was received and sent on to the Data Lake.

Chapter 16: Verifying the configuration

You must verify the configuration of your product with Infor OS.

Running this product from Infor Ming.le

- 1 Log into Infor Ming.le portal as a user. The user must belong to a role that is defined as an administrator in both Infor Ming.le and Mongoose.
 - a From the **App Menu**, click the icon for this product.
 - b If Single Sign On is configured, select the appropriate configuration and click **OK**.
The product session is opened.
 - c Open the Users form and verify this information:
User Name matches the Email address (Person ID) in Infor Ming.le.
Workstation Domain/ID matches the GUID in Infor Ming.le.
Roles you set in Infor Ming.le are set up as Groups for this user. You can change the user's groups here.
- 2 Log into Infor Ming.le as a non-administrative user who has access to both Infor Ming.le and the Mongoose-based application and perform these tasks:
 - Ensure that you can open any forms that the user has authorization to open, and that you cannot open any forms for which the user is not authorized.
 - Create a Homepage. Ensure that you can add a Mongoose widget on your Homepage and view a CloudSuite Industrial "home" page that you are authorized to open.
 - Ensure that you can view critical numbers that you are authorized to view in the In-Context KPI context app.

Verifying a drillback

Drillbacks are displayed in the Infor Ming.le user interface as links that open Mongoose to a specific screen filtered for specific results. For example, an Infor Ming.le search result, alert, or share can include a link to the form with a reference to specific Mongoose data.

To verify that the drillbacks are working:

- 1 Log in to Infor Ming.le portal as a user and open the Mongoose application.
- 2 Open a form that has a drillback, for example, Customers.
Customers is a form for a Mongoose-based application and not a native Mongoose form.
Note: The user interface does not indicate whether a screen has a drillback. In order for a screen to have a drillback, a drillback must be configured and a business context message must also be sent. See [Importing and configuring application drillbacks](#) on page 43 and [Understanding business context messages](#) on page 44.
- 3 Click the Share icon on the Infor Ming.le screen.
- 4 On the Share overlay, verify that the drillback icon is displayed, add a comment, and click **Share**.
- 5 In the Infor Ming.le feed, find the message that contains the drillback icon, and click the drillback to verify that it is working.

Appendix A: Troubleshooting

This section describes actions that you can perform to solve Infor OS integration issues.

Data is not flowing properly

Symptom

A problem is preventing the flow of BOD data to ION.

Possible solutions

After you complete the configuration setup, if BOD data is not flowing to ION, use ION OneView to identify the problem.

ION OneView can be found in the ION Desk app. You can perform these actions in ION OneView:

- Track business documents from a single consolidated view; search for documents using different search criteria.
- View all ION components that were triggered by the incoming document: Connection Points; document flow filters and content-based routing; mappings; ION engines for example, Monitor, Workflow, and Pulse; monitors; activation policies and workflows.
- View more details about these ION components:
 - Detailed properties for each ION component
 - List of events that were logged by each ION component while processing the message
 - Drilldown views that are used to display the appropriate management pages for the selected ION component
 - Visibility and correlation of the different messages sent or generated:
 - Original BOD messages that triggered the whole list of components displayed; Confirm BOD messages that were generated due to any error while processing the Original BOD message. Mapped/Updated BOD messages that were created during the processing of the Original BOD message
 - View content of BOD messages. This information is only available to authorized users, based on the roles and permissions that are configured.

For more information, see the section on ION OneView in the *ION Desk User Guide*.

To avoid business discrepancies between this product and integrated products, correct any errors as soon possible.

Purging BOD information

Symptom

Information from every inbound BOD, including sequencing information, is collected in a table in the Mongoose application. This data constantly builds up and, over time, cause performance issues.

Possible solutions

System administrators should periodically use the Purge BOD Information form to purge information from that table. You can set up a background task to perform this function automatically.

You can also use the Replication Document Inbox/Outbox Purge Utility to clear out older BODs from the Replication Document Inbox and Outbox.

Appendix B: Business events that generate outbound BODs

This table shows the events, which are user actions in Mongoose that generate an outbound BOD from Mongoose:

Verb	Noun	User action to generate the BOD
Process	AccountingBookDefinition	Run the Replication Document Manual Request Utility with AccountingBookDefinition selected. This should be done once per site.
Sync	AccountingChart	Run the Replication Document Manual Request Utility with AccountingChart selected. This should be done once per site, after the Chart of Accounts is set up at the site.
Sync	AccountingEntity	Run the Replication Document Manual Request Utility with AccountingEntity selected. This should be done once per site, after the tenant ID is set up at the site.
Process	AdvanceShipNotice	Perform material transactions that remove inventory tied to these types of orders: Project, Job, Customer Order, Purchase Order (with a negative quantity), RMA, Production Order or Transfer Order. Run the Multi-Site Quantity Move utility. Perform material transactions for a transfer order shipment, where the To warehouse is an external controlled warehouse and the To site is the local site.
Acknowledge	BillofMaterials	Acknowledge the receipt of a BillofMaterials from an external product.
Sync	BillofMaterials	Indicate that changes have been made in the application after receipt of a Process BillofMaterials from an external product
Acknowledge	BillToPartyMaster	Acknowledge the receipt of a BillToPartyMaster from an external product.

Business events that generate outbound BODs

Verb	Noun	User action to generate the BOD
Sync	BillToPartyMaster	Change any information on the Customers form. Change any information on the Ship Tos form.
Sync	Calendar	Add or update a shift calendar in Scheduling Shifts. Add a holiday. This creates a BOD that indicates all resources are busy on that day.
Sync	CarrierRoute	Select the Physically Shipped check box on the Special tab of the Delivery Orders form.
Sync	ChartOfAccounts	Create or update an account record in the Chart of Accounts form
Acknowledge	CodeDefinition	Acknowledge the receipt of a CodeDefinition from an external product

Verb	Noun	User action to generate the BOD
Sync	CodeDefinition	Education Major • Add an External Expense Type • Add an Indirect Labor Code • Add an Inventory Adjustment Reason Code • Add or update an ISO U/M on the Unit of Measure Codes form • Add an Item Category • Add an Obsolete Slow Moving Reason Code • Add an Opportunity Source • Add an Opportunity Stage • Add a Payment Type • Add a Position • Add a Price Code • Add a Product Code • Add a Project Cost Code • Add a Province/State • Add a Shift ID on the Scheduling Shifts form • Add a Ship Via Code • Add a Skill • Add a Tax Code of type Exempt • Add a Tax Jurisdiction • Add a Territory • Add a Training Course • Add a unit code on the Unit Code 1 - 4 forms • Add a Work Code • Add a Work Experience BODS for some other codes, such as language IDs and cost methods, are automatically generated when the Replication Document Manual Request Utility is run.
Process	ConstrainedResource	Insert or modify a Shift ID (creates a BOD for associated resources)
Acknowledge	ContactMaster	Acknowledge the receipt of a ContactMaster from an external product
Sync	ContactMaster	• Add or change a Contact • Create a Customer Sales Contact Cross-Reference
Sync	Contract	• Add or update Customer Order header • Add or update Customer Order Blanket Line

Verb	Noun	User action to generate the BOD
Sync	CreditTransfer	Create a Bank Reconciliations record for a vendor payment, employee payment, or customer payment
Sync	CurrencyExchangeRateMaster	Add or update a Currency Rate
Acknowledge	CustomerPartyMaster	Acknowledge the receipt of a CustomerPartyMaster from an external product
Sync	CustomerPartyMaster	- Change any information on the Customers form, Ship Tos form, or Customer Sales Contact Cross References form - Add a prospect on the Prospects form to send this BOD with a status of "Pending"
Sync	CustomerReturn	- Print the RMA Order Verification Report - Change the status of an RMA
Acknowledge	ExpenseReport	Acknowledge the receipt of an expense report.
Sync	FinancialCalendar	- Change any information on the Accounting Periods form - Run the change Reports To Entity utility
Sync	InventoryAdjustment	- Perform Quantity Adjustment - Perform Miscellaneous Receipt or Miscellaneous Issue - Perform Cycle Count Posting - one BOD is generated for each combination of item, warehouse, and location where the quantity on hand has changed - Perform Physical Inventory Posting - one BOD is generated for each combination of item, warehouse, and location where the quantity on hand has changed
Sync	InventoryCount	- Run the Cycle Count Posting utility - Run the Physical Inventory Posting utility One InventoryCount BOD is sent for each item/warehouse combination

Verb	Noun	User action to generate the BOD
Sync	Invoice	- Print and post an invoice through the Consolidated Invoicing form - Post an invoice through the Invoice Posting (A/R) form Note: The Invoice BOD publishes one line for every order line included in the Invoice. When Line Summarization is turned on in Consolidated Invoicing, the summary invoice lines are not published; instead, each customer order line will be published as a line in the Invoice BOD.
Acknowledge	ItemMaster	Acknowledge the receipt of a Process ItemMaster from an external product.
Sync	ItemMaster	- Create a new item on the Items form - Make a change to these fields on the Items form: Item, Description, ABC Code, Commodity, Cost Method, Material Status, Reason, Backflush, Last Change, U/M, Unit Cost, S/N Track, Lot Track, Source, Order Minimum, Order Maximum, Alternate Item - Add or change an item description on the Multi-Lingual Items form
Sync	Location	- Create a new Warehouse - Update General Parameters form (creates a Location BOD with the site as the location)
Sync	Opportunity	Create or update a record on the Opportunities, Customer Orders, or Opportunity Tasks form
Sync	PayableTracker	- Post a voucher or an Adjustment through the A/P Voucher Posting form - Post a payment from any of these forms: A/P Check Printing/Posting, A/P Draft Printing/Posting, A/P EFT Posting, A/P Wire Posting
Sync	PayableTransaction	- Post a voucher through the A/P Voucher Posting form - Post a payment from any of these forms: A/P Check Printing/Posting, A/P Draft Printing/Posting, A/P EFT Posting, A/P Wire Posting - Set or release a Hold status on a voucher
Acknowledge	PayFromPartyMaster	Acknowledge the receipt of a PayFromPartyMaster from an external product

Verb	Noun	User action to generate the BOD
Sync	PayFromPartyMaster	- Change any information on the Customers form - Change any information on the Ship Tos form
Sync	Person	- Add or update information about a salesperson or sales manager on the Salespersons form - Update information related to a salesperson on the Employees form (for employee sales people) or Vendors form (for outside sales people)
Sync	Personnel	Add or update information on these forms: Employee, Employee Skills, Employee Cert/License, Employee Education, Employee Training Courses, Employee Work Experience, or Employee Position
Acknowledge	PlanningSchedule	Acknowledge the receipt of a Planning Schedule from an external product.
Acknowledge	ProductionOrder	Acknowledge the receipt of a ProductionOrder from an external product
Sync	ProductionOrder	- Applied filters on the Background Task generate this BOD for a firm job that is scheduled for the first time. This does not apply to jobs that have just been processed by the TriggerProductionOrderBGSyncSp Background task. You can add subsequent filters to reduce the number of BODs triggered by this task. Without filtering, after every scheduling or planning function a significant number of BODs could be created. - Change the status of a job or production schedule. - Change the Qty Released of a job or production schedule when the status is Released. Note: Creating a job generates this BOD, but changes to a firm job do not generate this BOD. This BOD must be included in an initial load of ProductionOrder data.
Sync	ProjectMaster	Create a Project or change the project status.

Verb	Noun	User action to generate the BOD
Process	PulseAlert	One or more of these publications is triggered by a system event: • CustomerCreditHoldAlert • CustomerInteractionFollowupAlert • CustomerOrderCreditHoldAlert • CustomerShipmentAlert • DCShopFloorTransactionErrorAlert • JobMaterialCostAlert • JobOperationHoursAlert • JobOverProductionAlert • JobProjectedLateForCustomerAlert • LateOrderShippingAlert • LeadAssignedAlert • LockedUserAlert • OpportunityDueAlert • OverBudgetAlert • ProjectedLateShipAlert • ProspectInteractionFollowupAlert • ScheduledMaintenanceAddedAlert • ScheduledMaintenanceAlert • TransactionAmountAlert
Sync	PurchaseOrder	• Print the Purchase Order Report • Print the Builder Purchase Order Report • Change the Purchase Order status to Complete • Print a Change Order Report
Acknowledge	Quote	• Print the Purchase Order Report • Print the Builder Purchase Order Report • Change the Purchase Order status to Complete • Print a Change Order Report
Sync	Quote	Acknowledge the receipt of a Quote from an external product.
Sync	ReceivableTracker	Change the Estimate status to Quoted

Verb	Noun	User action to generate the BOD
Sync	ReceivableTransaction	- Generate an invoice for customers with an 'Open Item' balance method. The status is set as Open. - Generate a debit memo/finance charge for customers with an 'Open Item' balance method. The status is set as Open. - Make a payment and apply it to an invoice for customers with an 'Open Items' balance method. The status changes to Paid. - Generate a credit memo and attach it to an invoice for customers with an 'Open Item' balance method. This changes the status to Paid. (For bullet points 2, 3, and 4: If all or part of the debit memo/finance charge, payment, or credit memo is left unapplied, a second BOD is generated. Similar transactions are grouped and shown under an invoice number '0' with a status of Unapplied Cash.) - Create an invoice, payment, credit memo, debit memo or finance charge transaction for a customer with a 'Balance Forward' balance method. The system maintains one balance under invoice number '0'. Its status is set as follows: for a debit balance, the status is set as Paid; for a credit balance, the status is set as UnappliedCash.
Sync	ReceiveDelivery	- Perform material transactions that add inventory tied to these types of orders: Project, Job, Customer Order (with a negative quantity), Purchase Order, RMA, Production Order or Transfer Order that cause a material transaction to occur - Run the Multi-Site Quantity Move utility
Sync	RemittanceAdvice	Post a payment from any of these forms: A/P Check Printing/Posting, A/P Draft Printing/Posting, A/P EFT Posting, A/P Wire Posting, or Print/Post Payroll Checks
Sync	RemitToPartyMaster	- Add or update Vendor information - Update the vendor's address information - Update Bank Address information
Acknowledge	Requisition	Acknowledge receipt of a Requisition from an external product

Verb	Noun	User action to generate the BOD
Sync	Requisition	- Print the requisition on either the Purchase Requisition Report or the PO Requisition Report forms - Convert a requisition to a purchase order - Convert additional items on an existing requisition (in this case, another BOD is created, replacing the original BOD)
Acknowledge	SalesOrder	Acknowledge the receipt of a SalesOrder from an external product
Sync	SalesOrder	- Print the Customer Order Verification Report - Change order header status to Completed
Sync	SecurityRoleMaster	Add or delete a Group Authorization or change the group description.
Acknowledge	SecurityUserMaster	Acknowledge the receipt of a SecurityUserMaster from an external product.
Process	SecurityUserMaster	Add or change the user status, user description or workstation login.
Sync	ServiceOrder	Add or update Service Order or Service Order Operations or Lines.
Sync	ShipFromPartyMaster	- Add or update Vendor information - Update the vendor's address information - Update Bank Address information
Process	Shipment	- For customer orders or lines that are assigned to the local site and that are assigned to an external controlled warehouse: - Set the Customer Order Line status to Ordered. - Change U/M, due date, or quantity ordered on CO Line. - Customer Order or Customer related to the order is put on or taken off credit hold. - For transfer orders or lines where the Ship From site is the local site and the Ship From warehouse is an external controlled warehouse: - Create Transfer Order Line - Change U/M, schedule ship date, or quantity required on transfer order line

Business events that generate outbound BODs

Verb	Noun	User action to generate the BOD
Sync	Shipment	- Perform material transactions that remove inventory tied to these types of orders: Project, Job, Customer Order, Purchase Order (with a negative quantity), RMA, Production Order or Transfer Order - Run the Multi-Site Quantity Move utility
Acknowledge	ShpmentSchedule	Acknowledge the receipt of a ShipmentSchedule from an external product
Acknowledge	ShipToPartyMaster	Acknowledge the receipt of a ShipToPartyMaster from an external product
Sync	ShipToPartyMaster	Change any information on the Customers form, Ship Tos form, or Customer Sales Contact Cross References form
Sync	SourceSystemGLMovement	- Run the Change Reports To Entity utility - Run the Copy Balances to Budgets utility - Modify Chart of Accounts Budget and Plan information - Run the Rebalance Ledger Period Totals utility - Run Ledger Posting for Journals - Run the Mass Journal Posting utility
Sync	SourceSystemJournalEntry	Post journal entries to a ledger
Sync	SupplierPartyMaster	- Add or update Vendor information - Update the vendor's address information - Update Bank Address information
Sync	Transfer	- Perform a Transfer Order Ship - Perform a Transfer Order Receive - Perform a Combined Transfer Order Ship/Receive
Sync	WorkCenter	Add a Work Center

Appendix C: Inbound BOD usage

This table shows the incoming BODs that Mongoose can accept. It also indicates, when possible, what area of Mongoose shows the processed inbound data.

Verb	Noun	Destination of processed data in Mongoose
Load	BankStatement	Bank Statements form, and A/P automatic payments generated
Process	BillOfMaterials	Engineering Workbench, Current Operations, Engineering Board
Process	BillToPartyMaster	Customers form and related forms
Process	CodeDefinition	Depends on type of code
Process	ContactMaster	Contacts form and related forms
Process	CustomerPartyMaster	Customers form and related forms
Sync	EmployeeTimesheet	Service order transactions, attendance, project Labor transactions, unposted job transactions (Indirect, Run or Setup)
Sync	ExpenseReport	ptrx table
Sync	InventoryAdjustment	Miscellaneous Receipt and Miscellaneous Issue forms
Sync	InventoryCount	Miscellaneous Receipt and Miscellaneous Issue forms
Process	ItemMaster	Items form and related forms
Process	PayFromPartyMaster	custaddr table
Sync	PlanningSchedule	coitem table (blanket items)
Process	ProductionOrder	job table
Process	Quote	Estimates form and related forms
Acknowledge	PulseAlert	N/A

Verb	Noun	Destination of processed data in Mongoose
Sync	ReceiveDelivery	If Type="Transfer" and Status="Received" create transfer order receipt. If Type="Purchase Order" and Status="Received" create PO receipt If Type="Customer Return" create RMA receipt If Type="Sales Order" create negative CO receipt.
Process	Requisition	preq table
Process	SalesOrder	Customer Orders form and related forms
Sync	SecurityUserMaster	coitem or trnitem table (depending on record type), last_external_shipment_doc_id
Acnowledge	Shipment	If Type="Transfer" and Status="Shipped" create Transfer Order Shipment If Type="Purchase Order" create Purchase Order Return If Type="Sales Order" and Status="Shipped" create Customer Order Shipment
Sync	Shipment	coitem or trnitem table (depending on record type), last_external_shipment_doc_id
Sync	ShipmentSchedule	coitem table (blanket items)
Process	ShipToPartyMaster	Customer Ship Tos form and related forms

Appendix D: BODs used in integrations with this product

This section contains the list of BODs that are available with Mongoose.

Outbound BODs from Mongoose to integrated products

This table shows the BODs that are available with Mongoose. Where the product is blank, the BOD is not currently used by interfacing products, but the BOD is generated by Mongoose and is available to be processed through ION and any product that is set up to receive it.

Verb	Noun	To products
Process	AccountingBookDefinition	
Sync	AccountingChart	Infor Local.ly Reporting
Sync	AccountingEntity	Infor Local.ly Reporting
Process	AdvanceShipNotice	SCE Warehouse Management
Acknowledge	BankStatement	Infor Local.ly GEMS
Acknowledge	BillOfMaterials	PLM Accelerate
Acknowledge	BillToPartyMaster	Infor Back Office Connect
Sync	BillToPartyMaster	CRM e-Commerce Infor Back Office Connect SCE Warehouse Management
Sync	Calendar	
Sync	CarrierRoute	Automotive Exchange
Sync	ChartOfAccounts	Infor Local.ly Reporting
Acknowledge	CodeDefinition	

Verb	Noun	To products
Sync	CodeDefinition	CRM Expense Management Infor Back Office Connect PLM Accelerate Infor Local.ly Reporting
Process	ConstrainedResource	
Acknowledge	ContactMaster	
Sync	ContactMaster	CRM Infor Back Office Connect
Sync	Contract	Automotive Exchange
Sync	CreditTransfer	
Sync	CurrencyExchangeRateMaster	Expense Management
Acknowledge	CustomerPartyMaster	
Process	CustomerPartyMaster	Infor Back Office Connect
Sync	CustomerPartyMaster	CRM e-Commerce Infor Back Office Connect Infor Local.ly Reporting PLM Accelerate SCE Warehouse Management
Sync	CustomerReturn	CRM Infor Back Office Connect
Sync	DebitTransfer	Infor Local.ly GEMS
Sync	FinancialCalendar	Infor Local.ly Reporting
Sync	InventoryAdjustment	
Sync	InventoryCount	
Sync	Invoice	Automotive Exchange CRM e-Commerce Infor Back Office Connect Infor Local.ly Reporting
Acknowledge	ItemMaster	PLM Accelerate

Verb	Noun	To products
Sync	ItemMaster	CRM e-Commerce Infor Back Office Connect PLM Accelerate SCE Warehouse Management
Sync	Location	CRM (ICBOE) e-Commerce
Sync	Opportunity	
Sync	PayableTracker	
Sync	PayableTransaction	
Acknowledge	PayFromPartyMaster	Infor Back Office Connect
Sync	PayFromPartyMaster	CRM Infor Back Office Connect
Sync	Person	
Sync	Personnel	Expense Management
Acknowledge	PlanningSchedule	
Acknowledge	ProductionOrder	
Sync	ProductionOrder	Expense Management
Sync	ProjectMaster	Expense Management
Process	PulseAlert	Infor Ming.le
Sync	PurchaseOrder	EAM SCE Warehouse Management
Acknowledge	Quote	
Sync	Quote	CRM Infor Back Office Connect
Sync	ReceivableTracker	
Sync	ReceivableTransaction	CRM Infor Back Office Connect Infor Local.ly Reporting
Sync	ReceiveDelivery	EAM
Sync	RemittanceAdvice	Expense Management
Sync	RemitToPartyMaster	

Verb	Noun	To products
Acknowledge	Requisition	EAM
Sync	Requisition	
Acknowledge	SalesOrder	
Sync	SalesOrder	CRM e-Commerce Infor Back Office Connect
Sync	SecurityRoleMaster	Infor Ming.le
Process	SecurityUserMaster	Infor Ming.le
Sync	ServiceOrder	Expense Management
Sync	ShipFromPartyMaster	
Process	Shipment	SCE Warehouse Management
Sync	Shipment	Auto Exchange CRM e-Commerce Infor Back Office Connect
Acknowledge	ShipToPartyMaster	
Sync	ShipToPartyMaster	e-Commerce CRM Infor Back Office Connect SCE Warehouse Management
Sync	SourceSystemGLMovement	Infor Local.ly Reporting
Sync	SourceSystemJournalEntry	Infor Local.ly Reporting
Sync	SupplierPartyMaster	EAM PLM Accelerate SCE Warehouse Management
Sync	Transfer	
Sync	WorkCenter	PLM Accelerator

Inbound BODs to Mongoose from integrated products

This table shows the BODs that can be received and processed by Mongoose. Where the product is blank, the BOD is not currently used by interfacing products.

Verb	Noun	From products
Process	BankStatement	Infor Local.ly GEMS
Process	BillOfMaterials	PLM Accelerate
Process	BillToPartyMaster	CRM Infor Back Office Connect
Process	CodeDefinition	Infor Back Office Connect
Process	ContactMaster	CRM Infor Back Office Connect
Process	CustomerPartyMaster	CRM Infor Back Office Connect Motion RW
Sync	EmployeeTimesheet	Expense Management
Sync	ExpenseReport	Expense Management
Sync	InventoryAdjustment	SCE Warehouse Management
Sync	InventoryCount	SCE Warehouse Management
Process	ItemMaster	PLM Accelerate
Process	PayFromPartyMaster	
Process	PlanningSchedule	Automotive Exchange
Process	ProductionOrder	
Acknowledge	PulseAlert	Infor Ming.le
Process	Quote	CRM
Sync	ReceiveDelivery	SCE Warehouse Management
Process	Requisition	EAM
Process	SalesOrder	CRM e-Commerce
Sync	SecurityUserMaster	Infor Ming.le
Acknowledge	Shipment	SCE Warehouse Management
Sync	Shipment	SCE Warehouse Management
Sync	ShipmentSchedule	

BODs used in integrations with this product

Verb	Noun	From products
Process	ShipToPartyMaster	CRM Infor Back Office Connect
